Depermissioning Web3: a *P*ermissionless *A*ccountable *R*PC *P*rotocol for Blockchain Networks

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DistriNet, KU Leuven

KU LEUVEN DistriNet

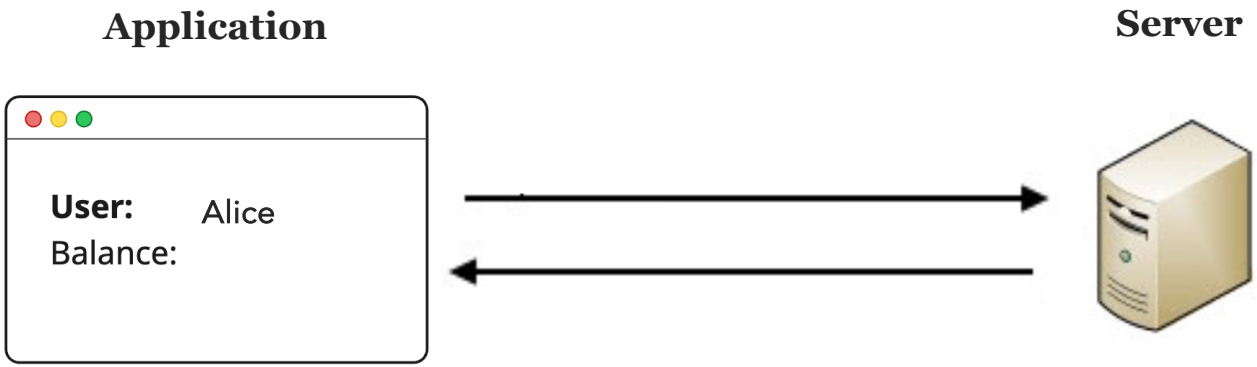
The Web3 Vision: Verifiable and Decentralized

verifiable

middleman-free

ownership

Web2: Your server owns the state



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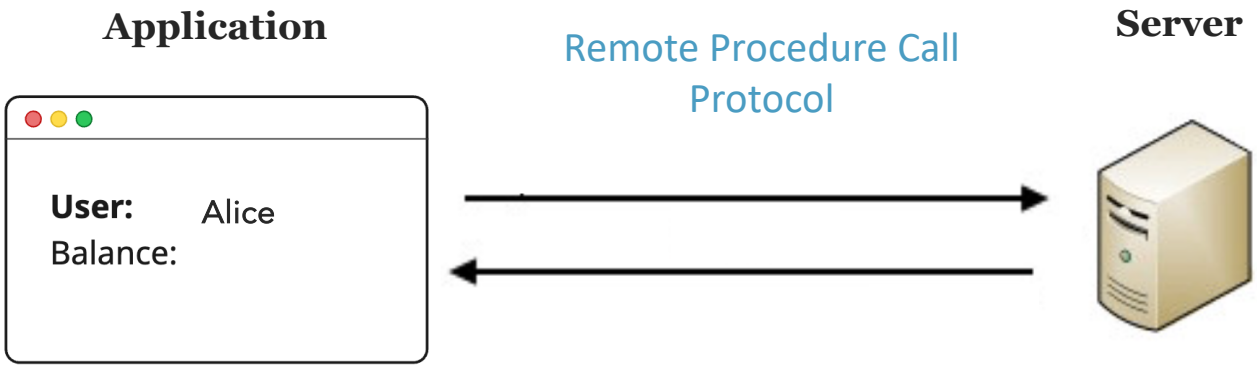

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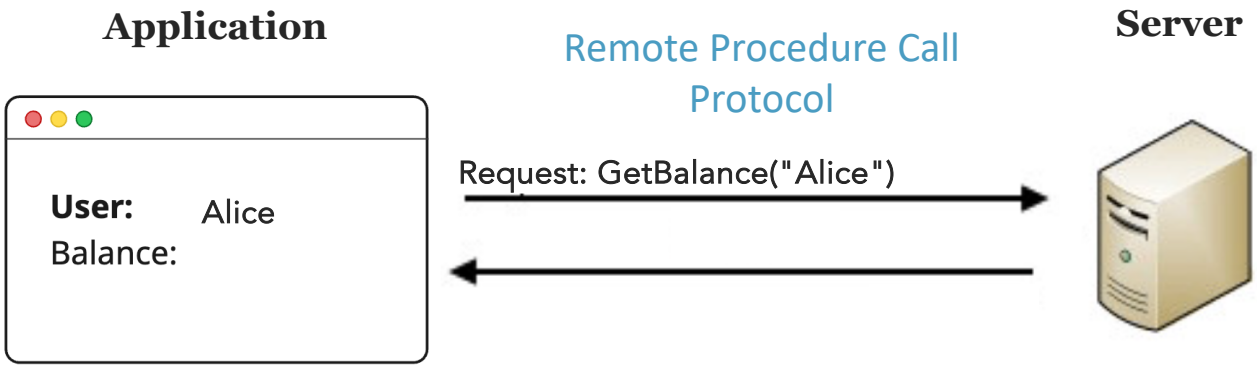

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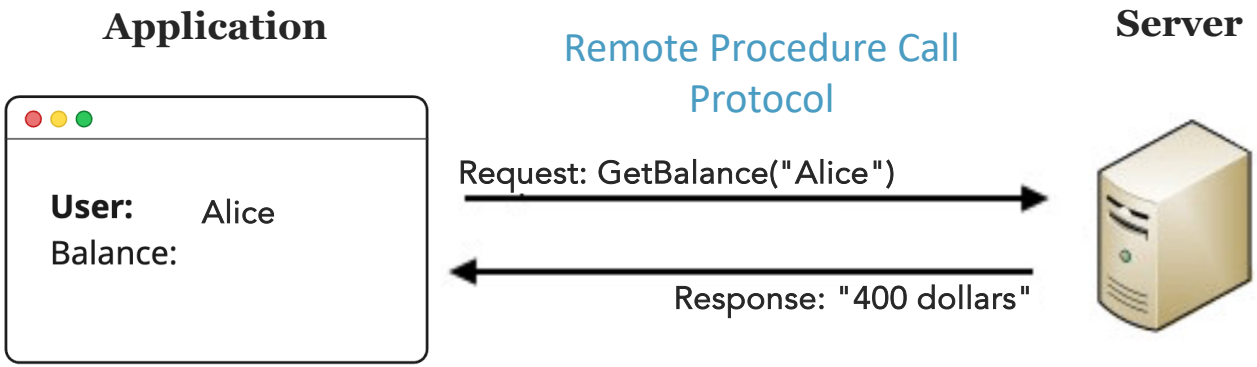

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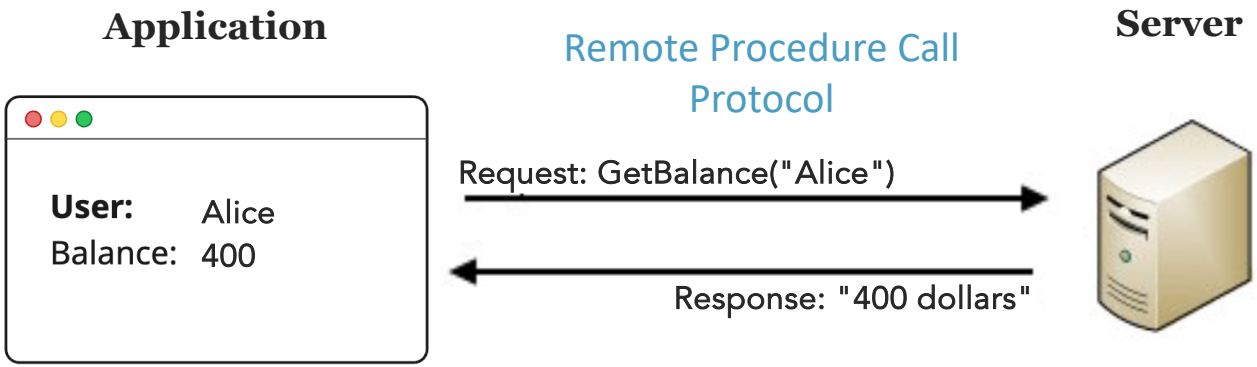
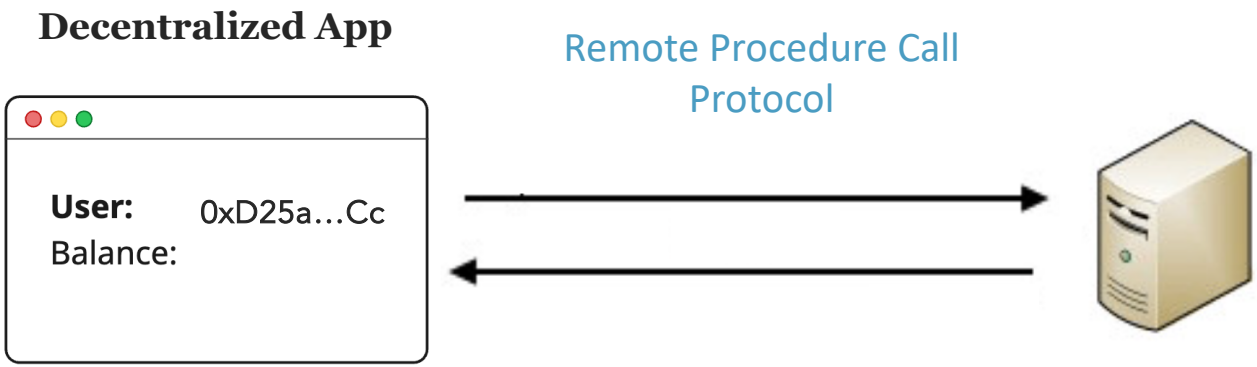


Web3: Your node mirrors the state of the chain



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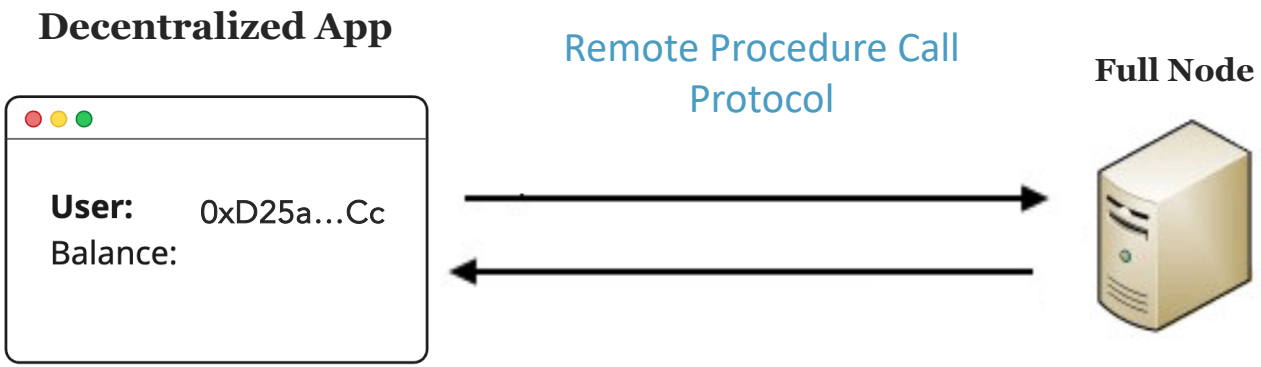

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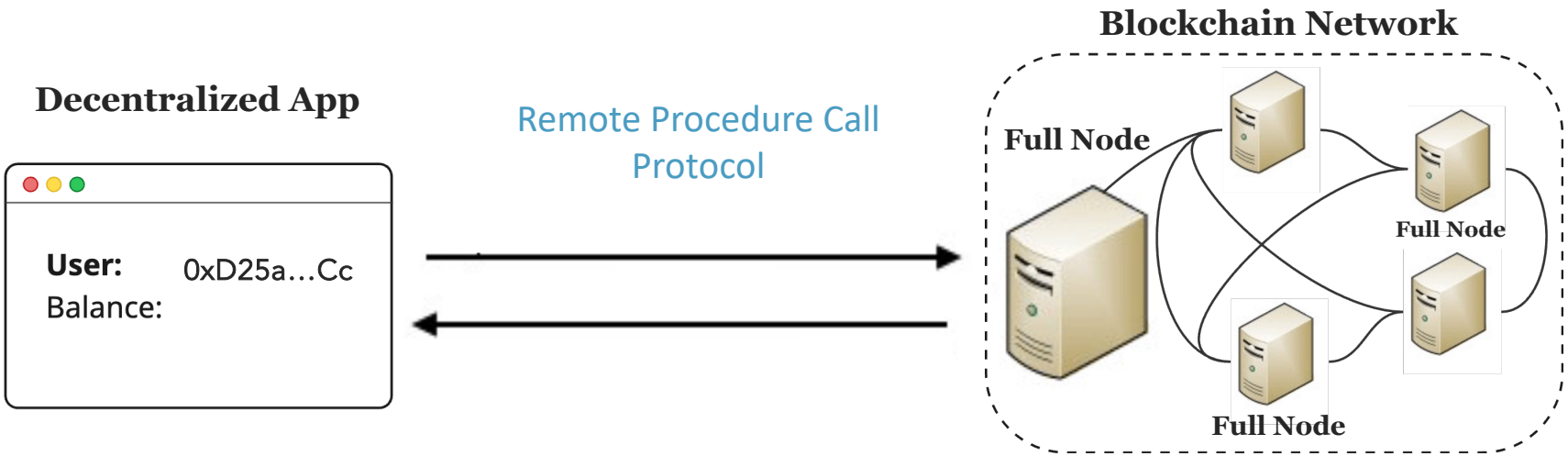

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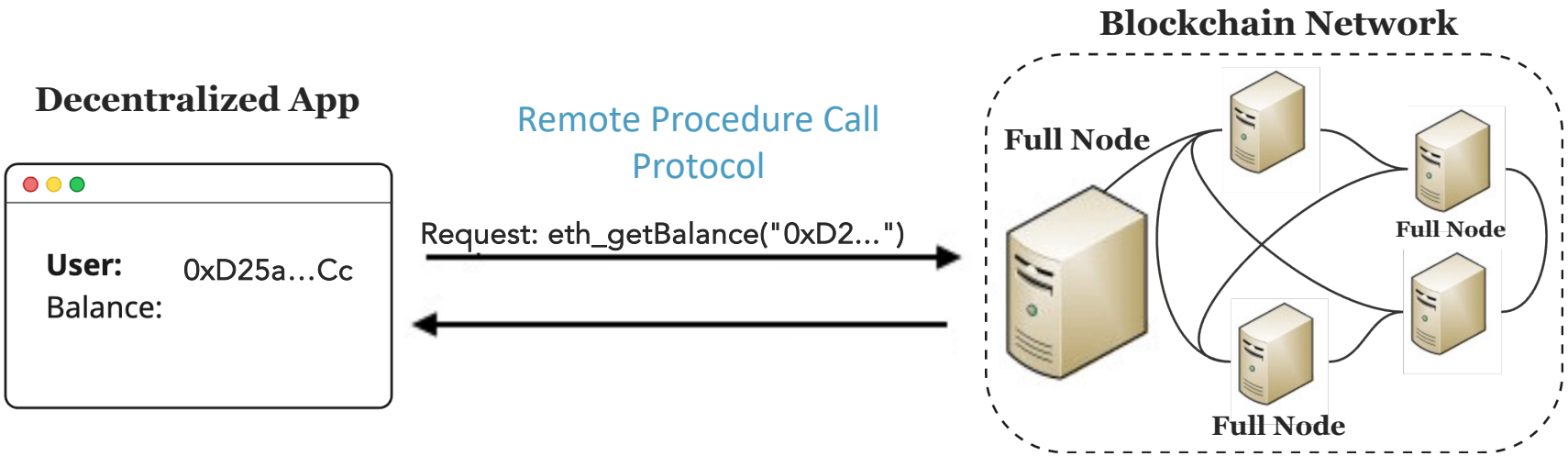

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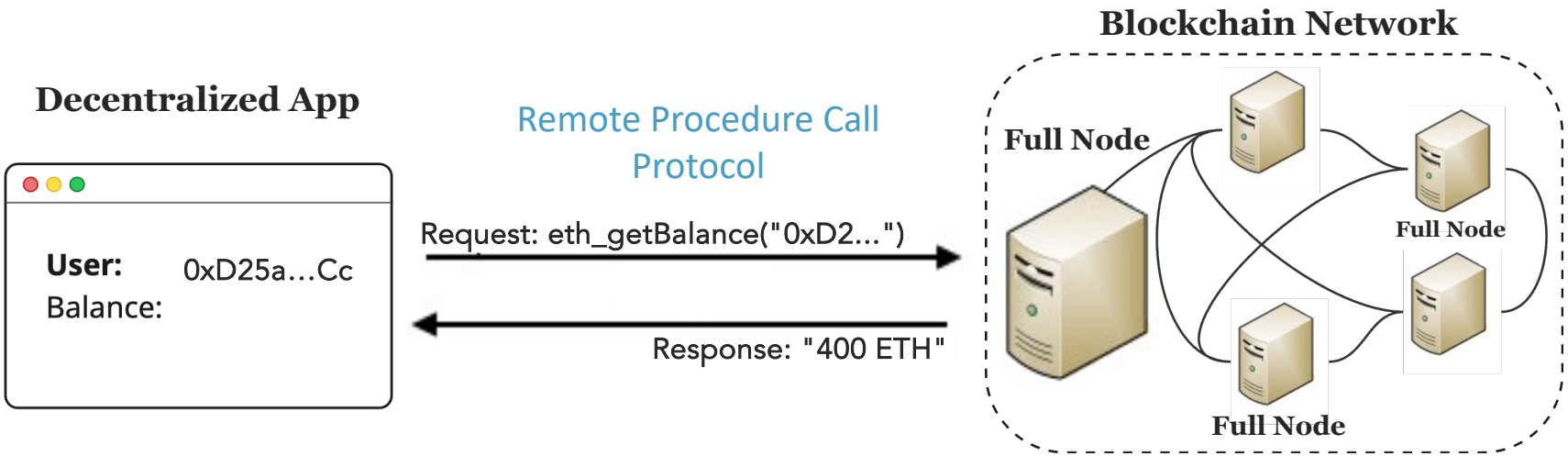

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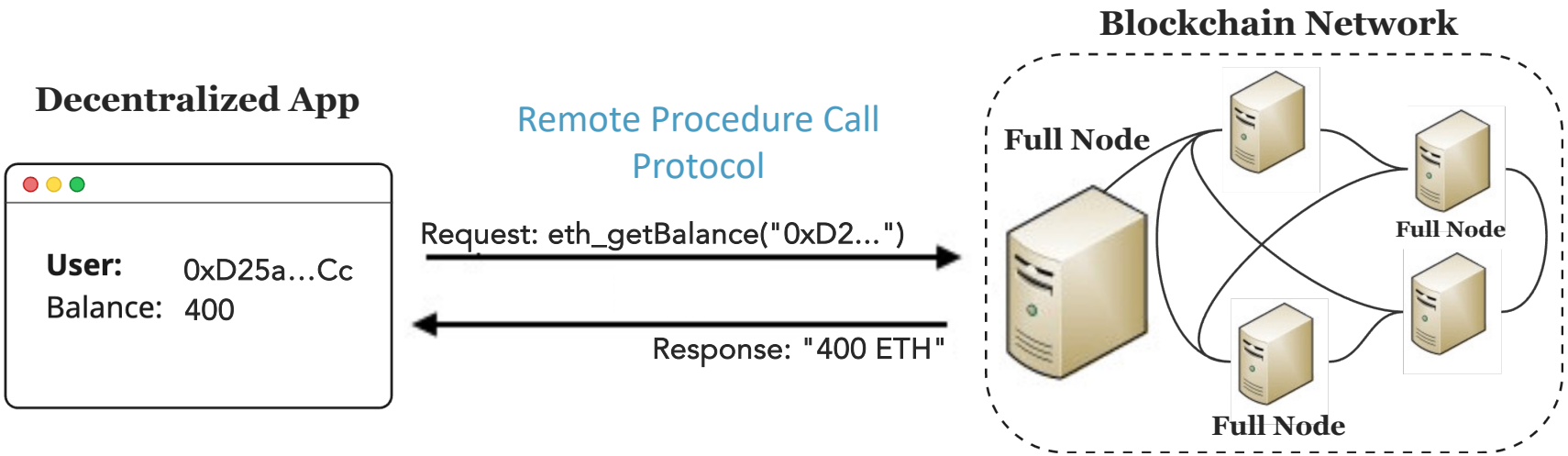

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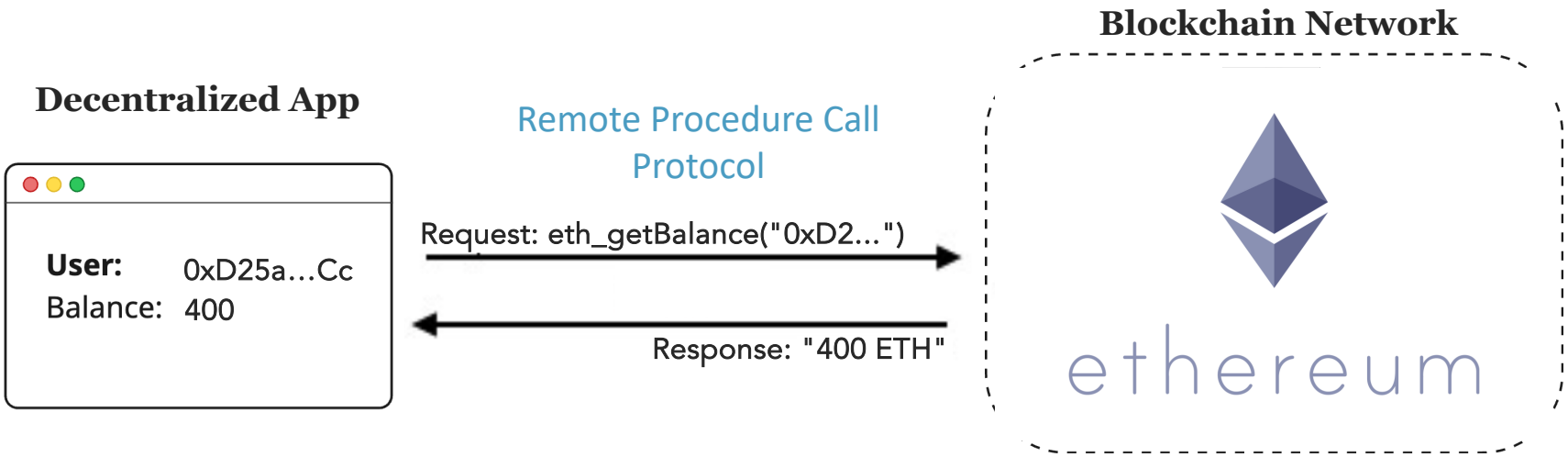

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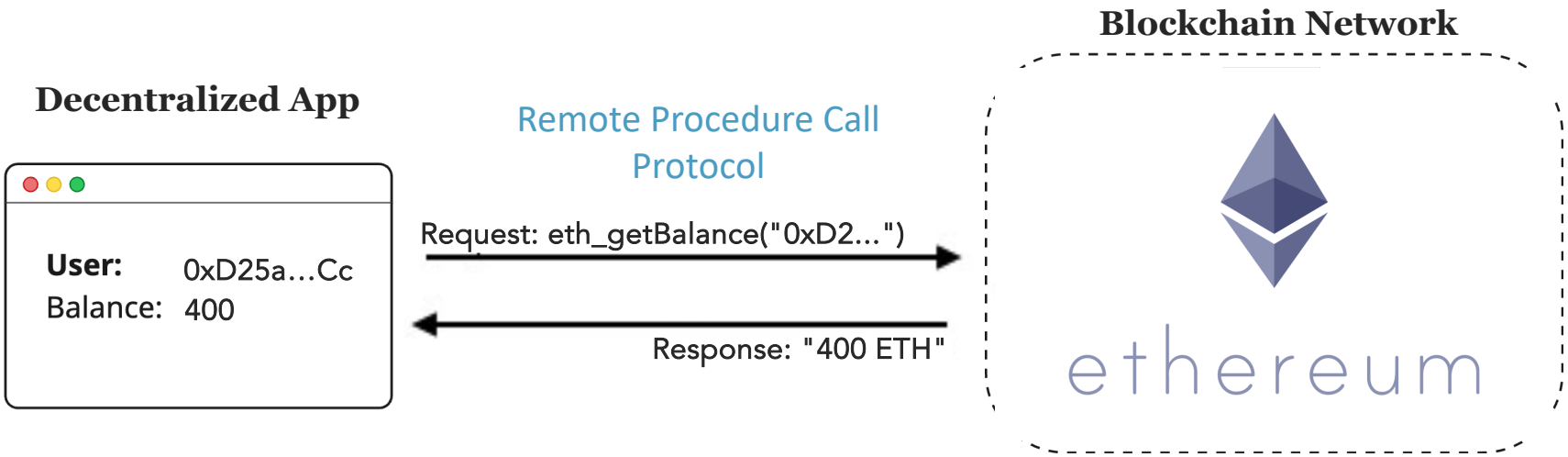

One of the biggest web3 networks

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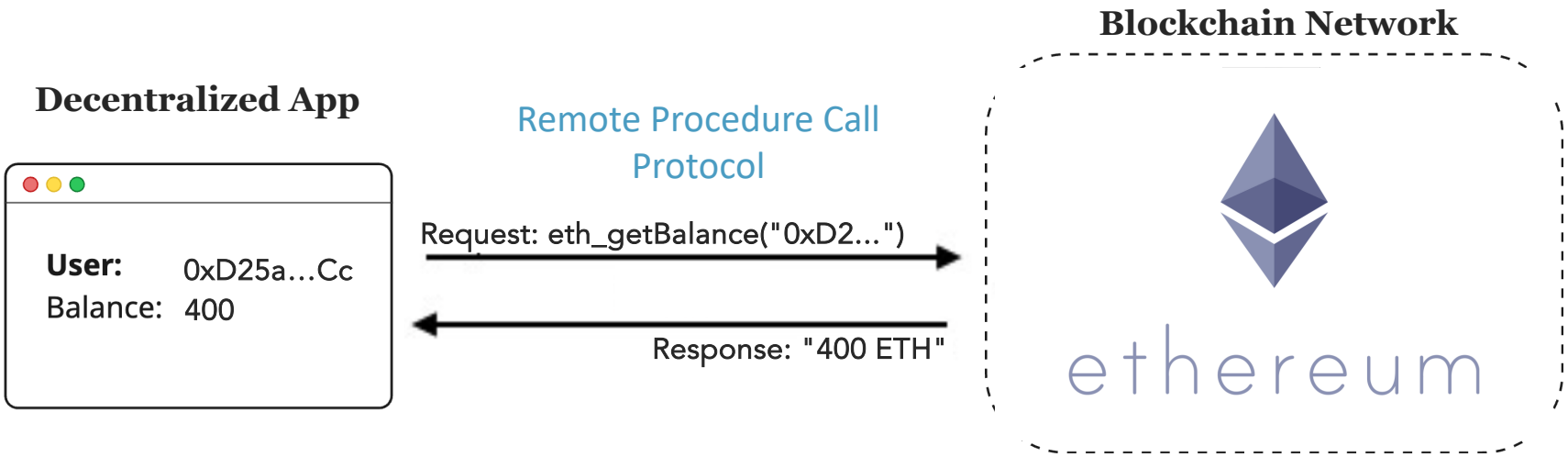
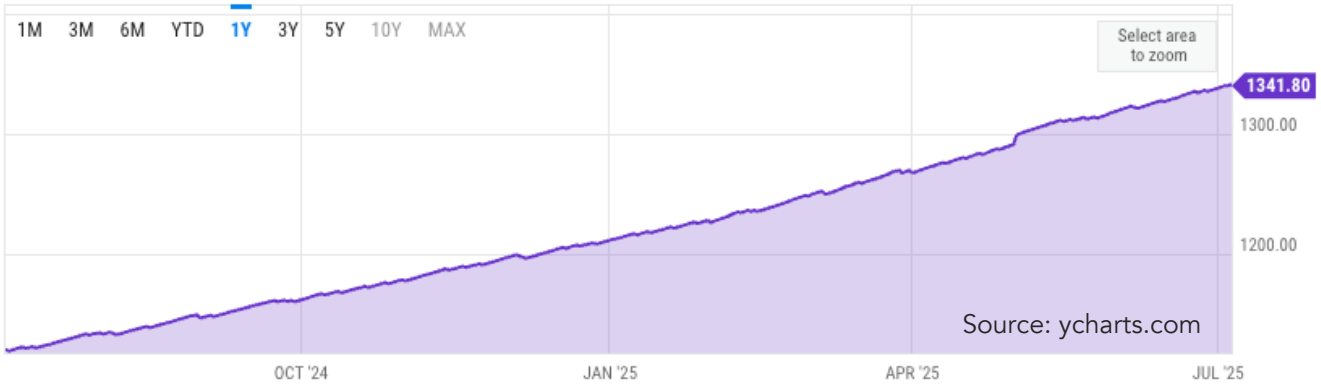


One of the biggest web3 networks

Enable smart contracts and dApps

Running an Ethereum full node costs a lot!



Ethereum Chain Full Sync Data Size

for 5 July 2025

increase from one year ago

Many use Ethereum, but few run nodes

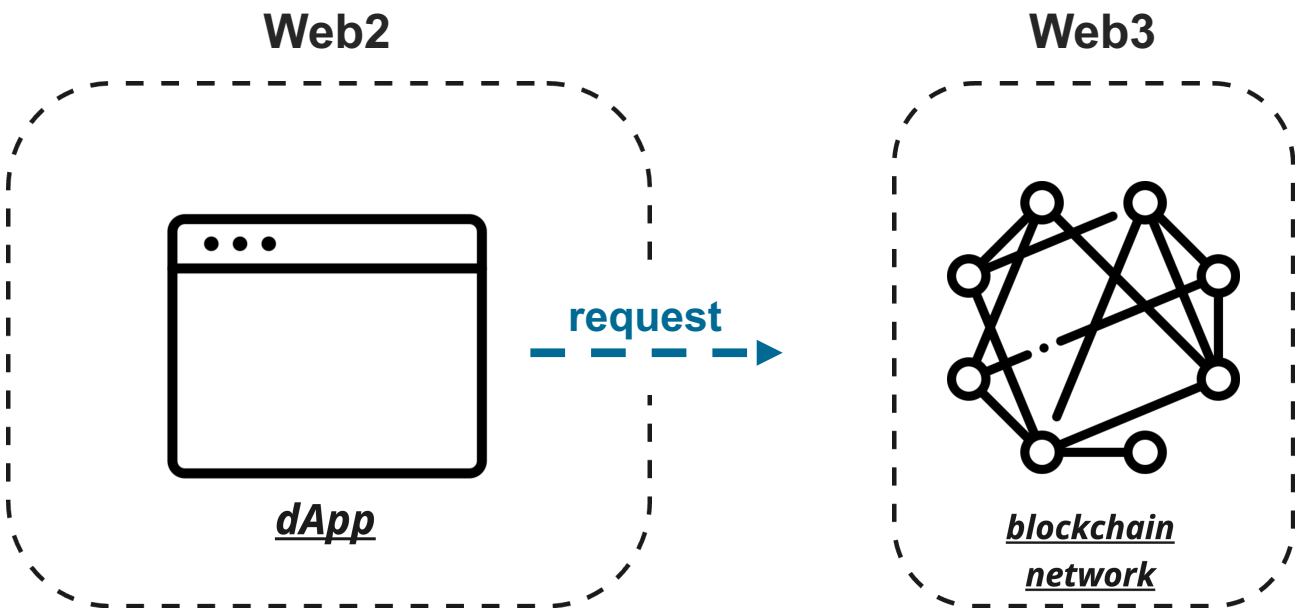
12,244

Number of Ethereum Mainnet Nodes [1]

› Ethereum Daily Active Addresses is at a current level of **400,000**.

[1] Source of statistics: etherscan.io, collected on 6 July 2025

Web3 serving layer depends on trusted intermediaries



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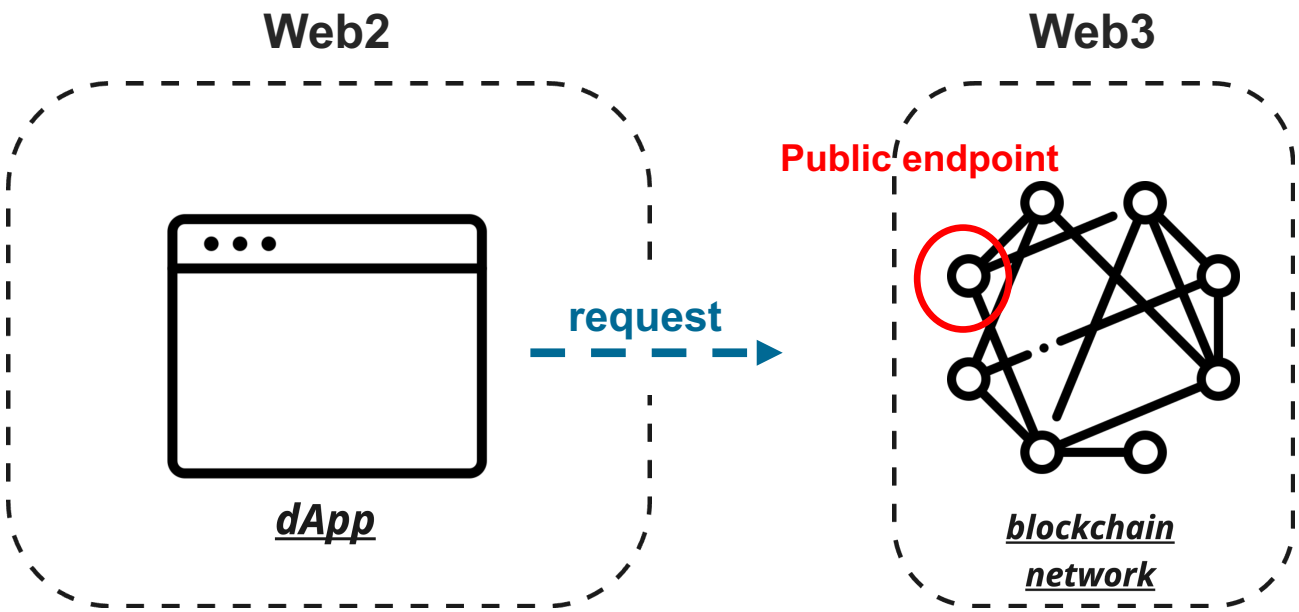
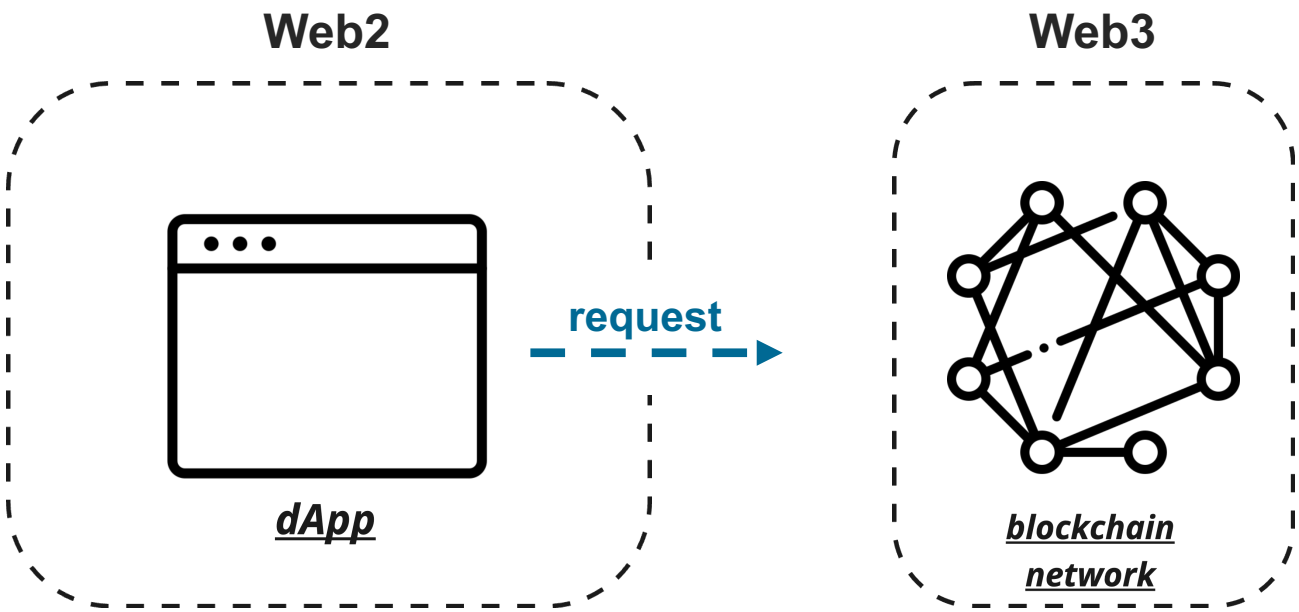
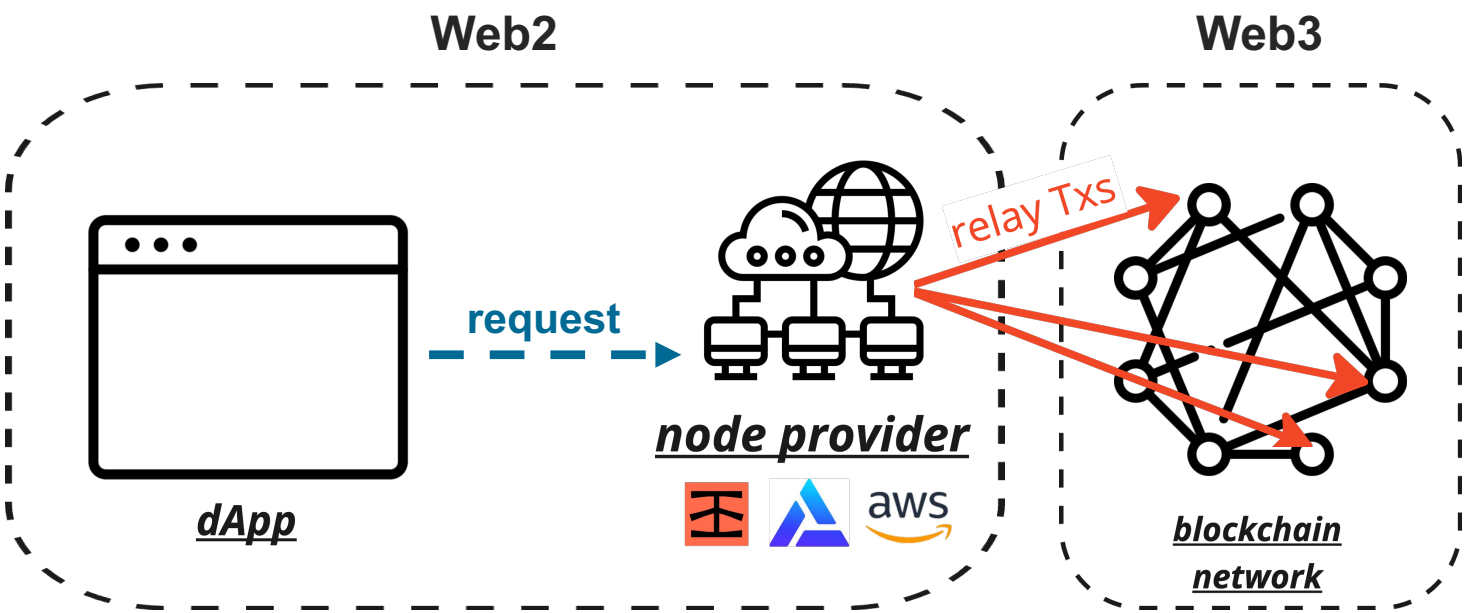



Web2
Web3
request
dApp
blockchain
network


Web2
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request
relay Txs
dApp
node provider
aws
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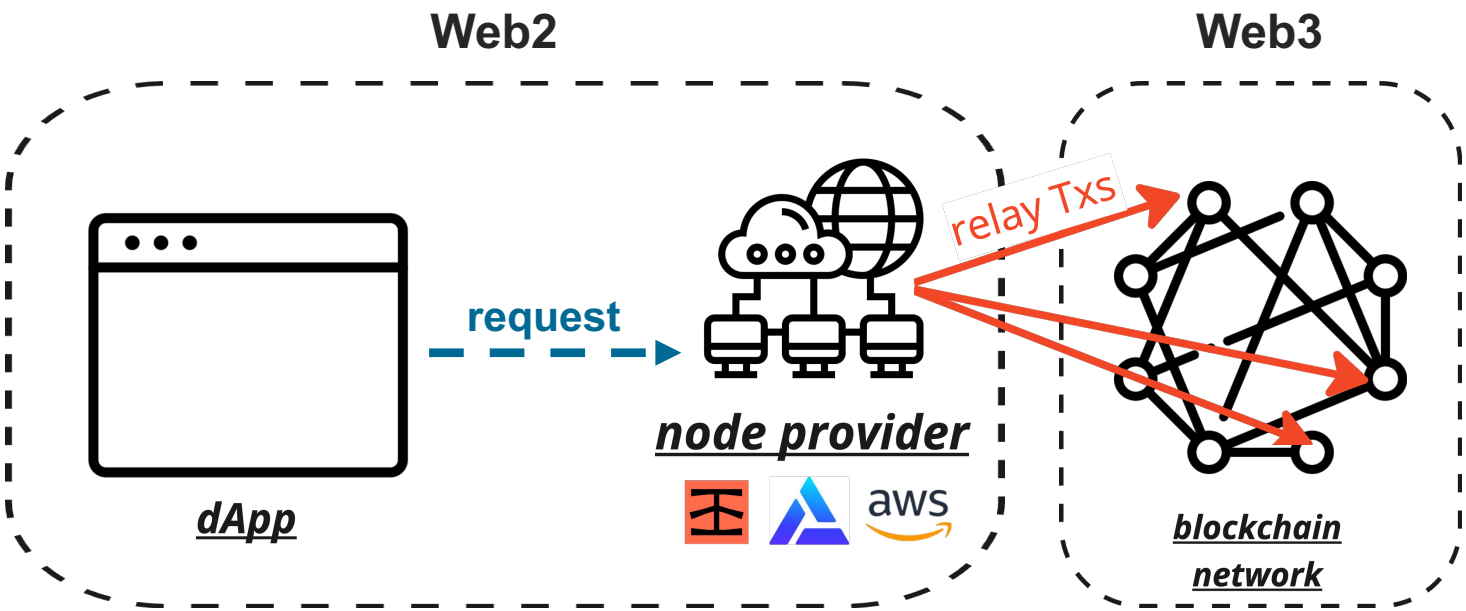
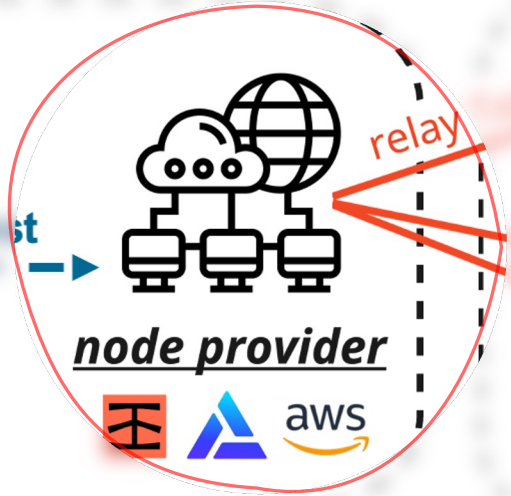


Web3 serving layer depends on trusted intermediaries

Node Provider

- Host a full node and answer requests
- Node-as-A-Service
- Registration required



Confidential information involved when registering

- Registration process from ***5 node providers*** (Infura has over 400,000 users.)

Node providers list
Infura
Alchemy
Ankr
Quicknode
Chainstack

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- Payment: 3 / 5 only supports credit cards

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Web3 access still starts with Web2 sign-ups

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weihong.wang@kuleuven.be
PaSsW0Rd

→ https://mainnet.infura.io/v3/**<API-KEY>**

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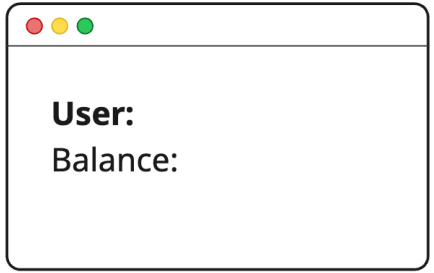
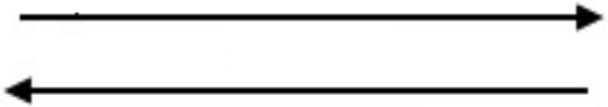
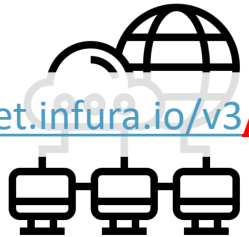
Decentralized App







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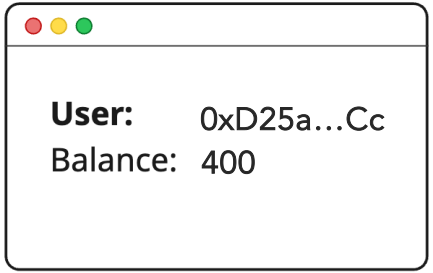
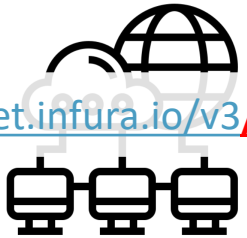


Request: eth_getBalance("0xD2...")

Response: "400 ETH"



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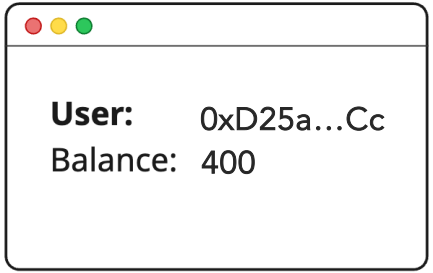

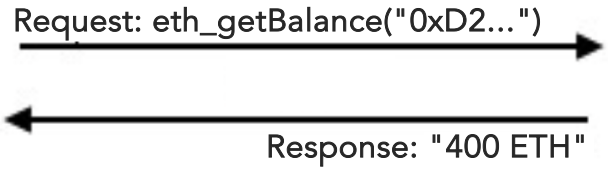

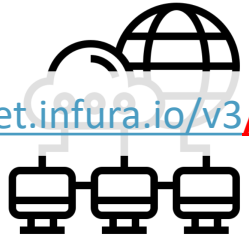

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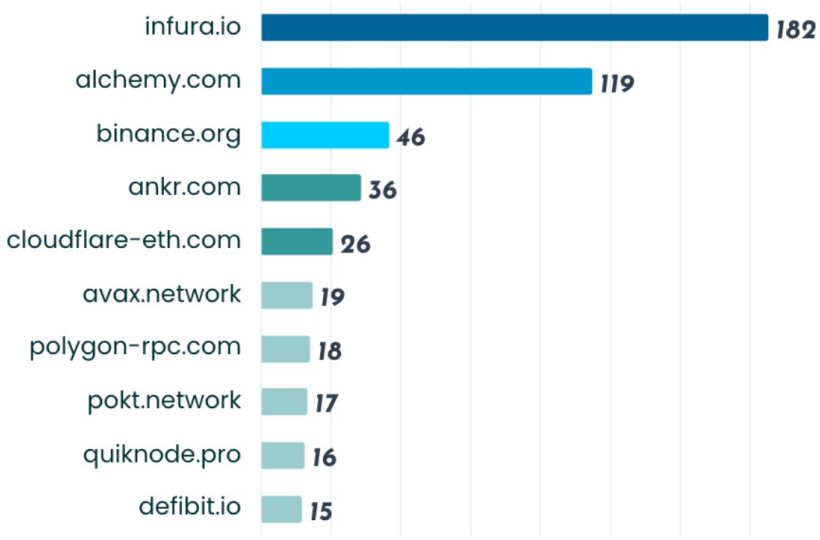
node provider

Requests in total: 400,000

2 USD

A few node providers see most Web3 traffic



Number of DApps Using Each Node Provider

Note: a single Dapp may send traffic to multiple node providers.

- Dataset: Web traffic from 1572 dApps*
- We focused on ***383*** dApps that send JSON-RPC calls from the front-end.

*Christof Ferreira Torres, Fiona Willi, and Shweta Shinde. 2023. Is your wallet snitching on you? an analysis on the privacy implications of web3. (Usenix Security '23)

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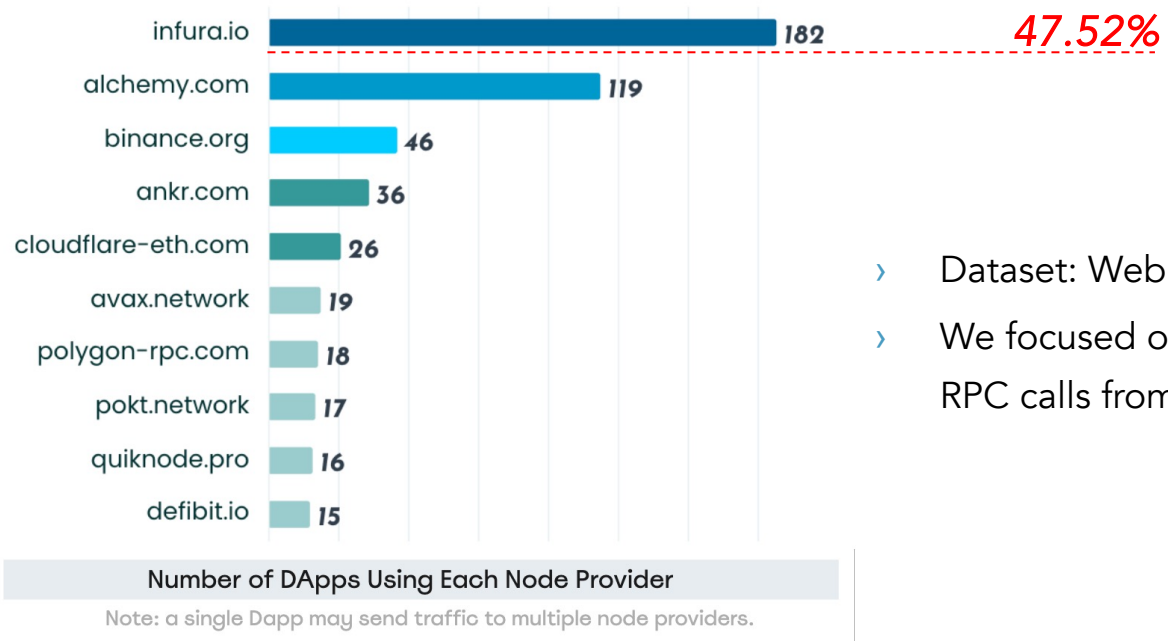

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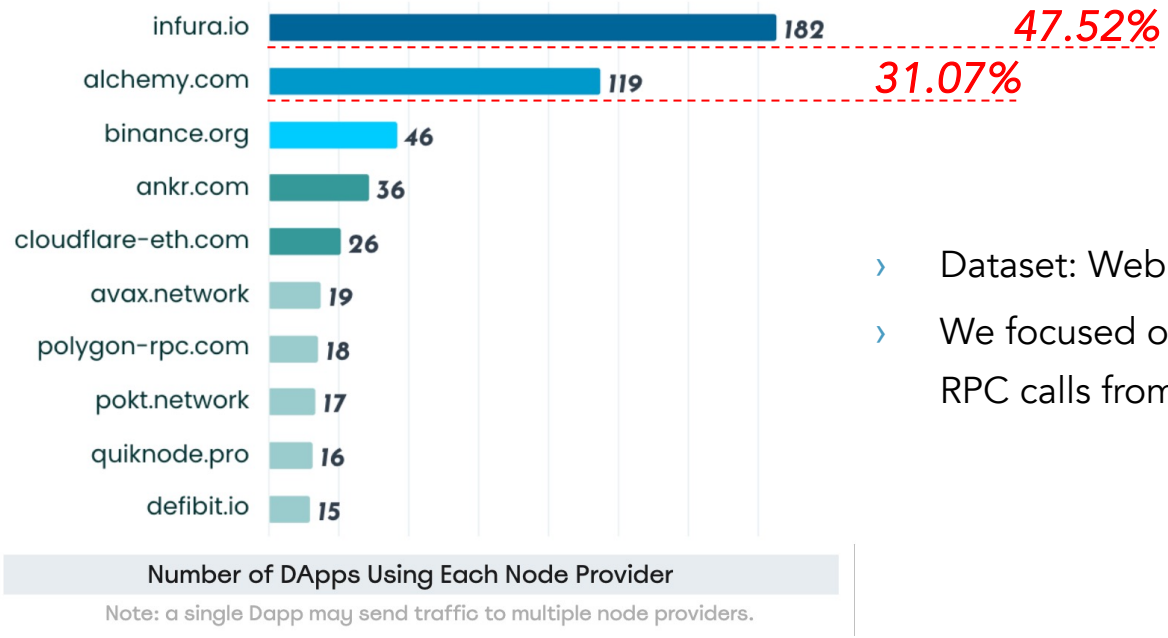

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Node Providers can see, block, or break access

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https://cointelegraph.com › news › infura-is-to-blame-f...

Infura is to blame for MetaMask's violation of the crypto spirit

13 Feb 2023 — ConsenSys announced in November that **Infura, MetaMask's API provider**, was collecting wallet and IP addresses from MetaMask's users.

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09 Aug 2022

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09 Aug 2022

Decrypt
https://decrypt.co › metamask-ethereum-apps-down-inf...

MetaMask, Ethereum Apps Down as Infura Suffers Outage

22 Apr 2022 — That's because the widely used infrastructure service **Infura is suffering an outage**, which is impacting an array of apps and services built on ...

Key Properties for a Decentralized Serving Layer

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1. No prior trust or sign-up

Permissionless

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2. Fraudulent behavior can be caught

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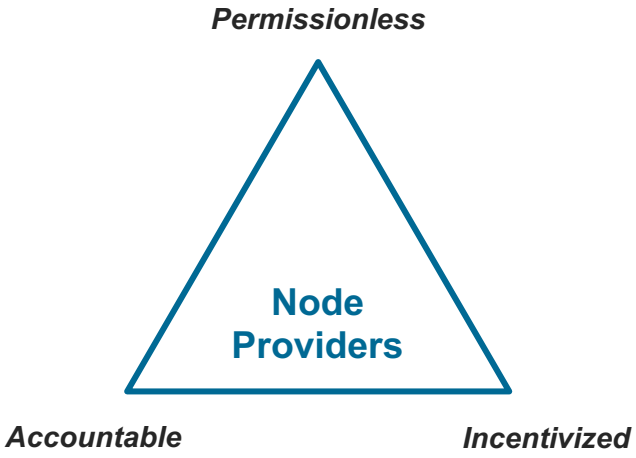
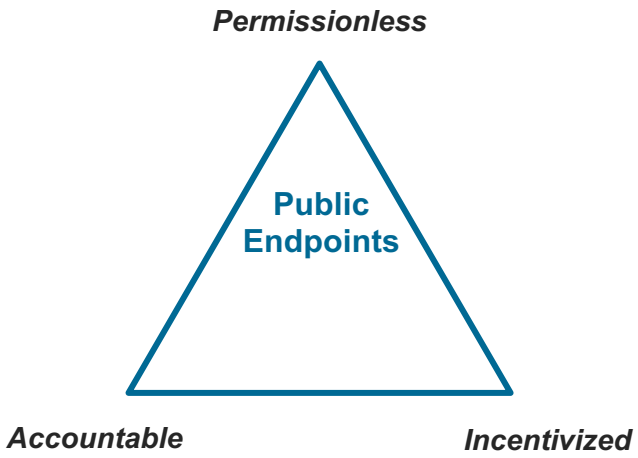
3. Fair monetary compensation

Existing Solutions Highlight the Access Dilemma





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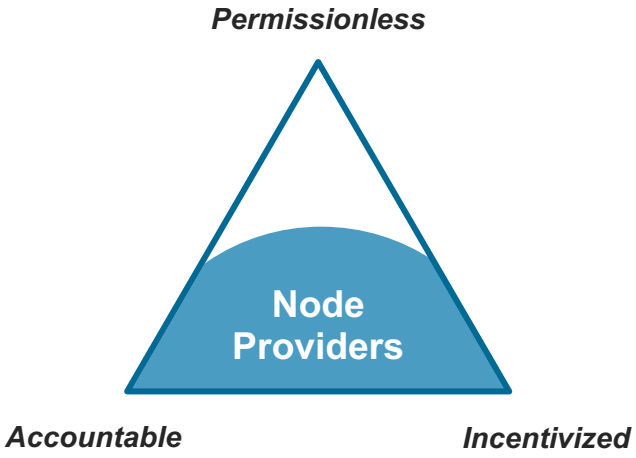

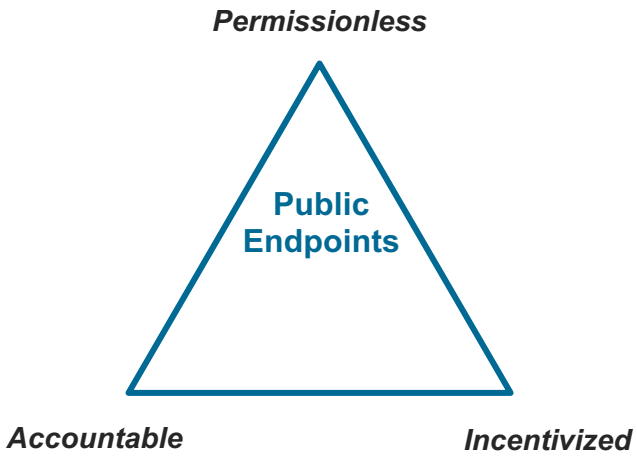

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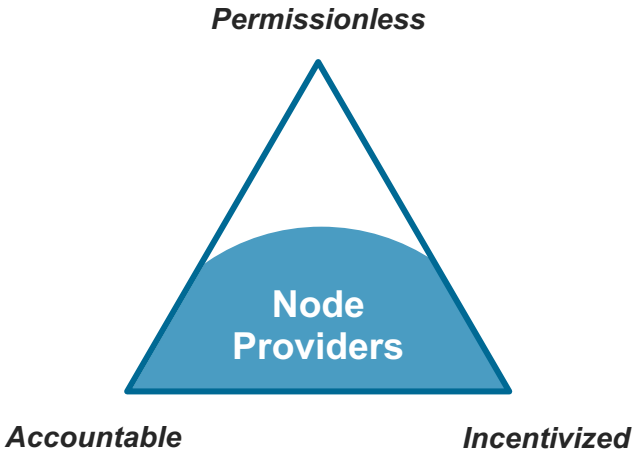

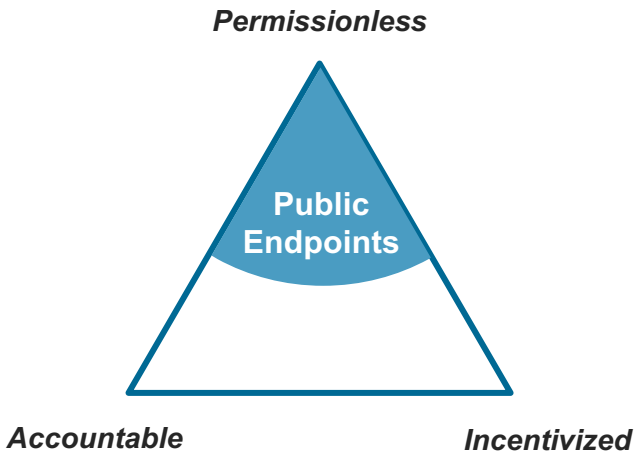

Consensus is Solved

Proof of Work / Stake

Block mining rewards and tx fees

Misbehavior causes slashing

For miners and validators

Consensus is Solved — Access isn't

Proof of Work / Stake

Block mining rewards and tx fees

Misbehavior causes slashing

For miners and validators

No standard protocol

No incentives

Trust-based access

For nodes that serve data

Contribution – A *P*ermissionless *A*ccountable *R*PC *P*rotocol

1. No prior trust or sign-up
2. Fraudulent behavior can be caught
3. Fair monetary compensation

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4. Implementation and evaluation in Ethereum

Challenges and Solutions

Main problems with the current node provider

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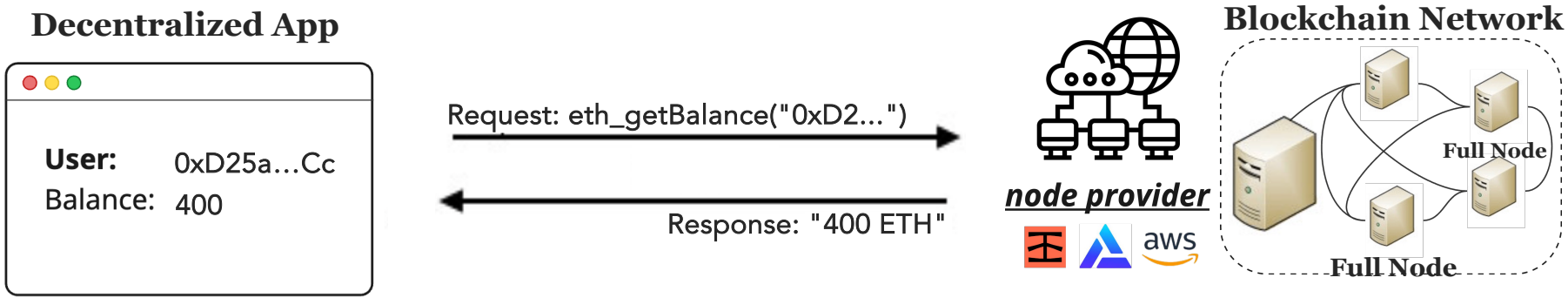

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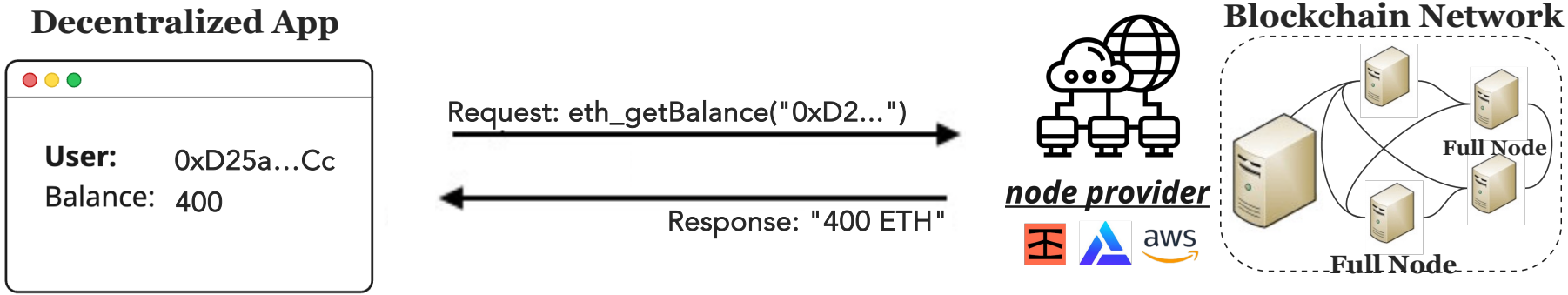
weihong.wang@kuleuven.be
PaSsW0Rd

https://mainnet.infura.io/v3/<API-KEY>



Requests in total: 400,000 2 USD

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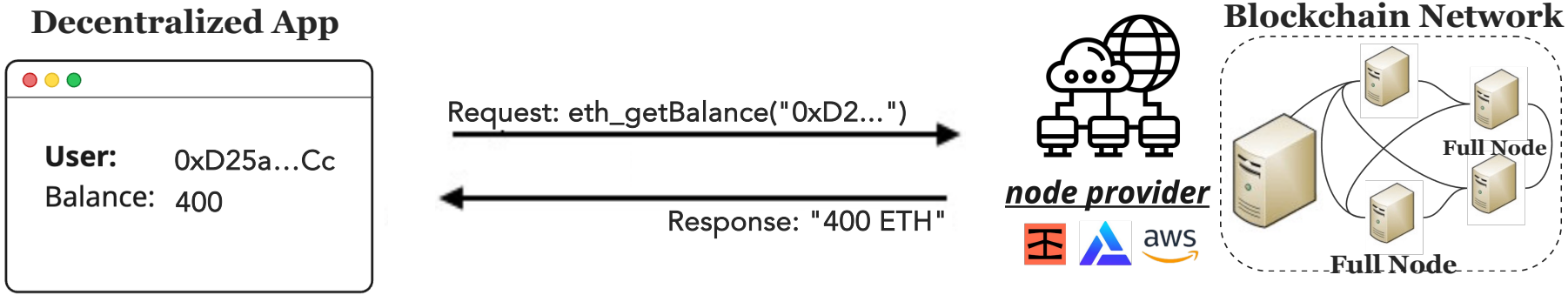

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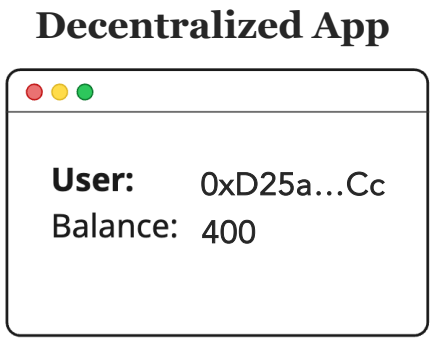

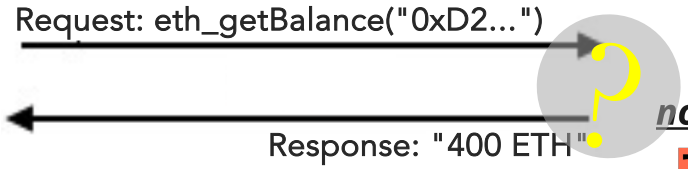

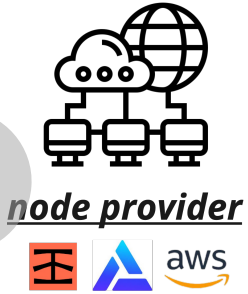

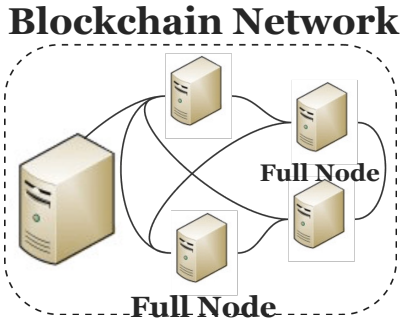

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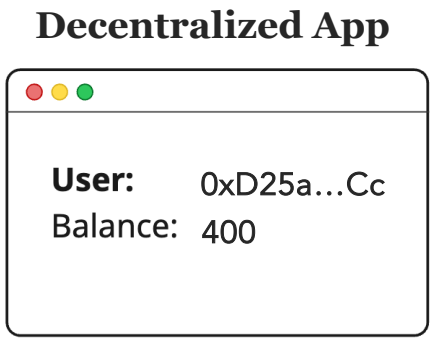
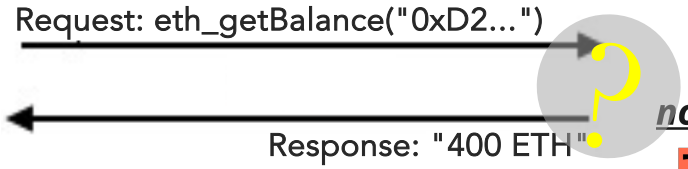
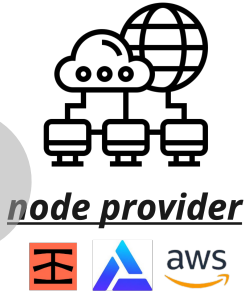
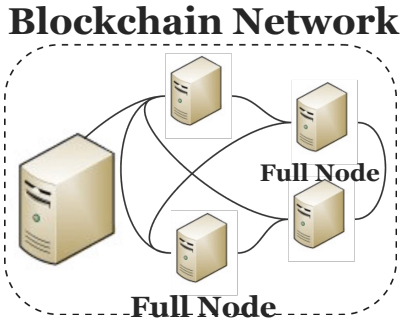




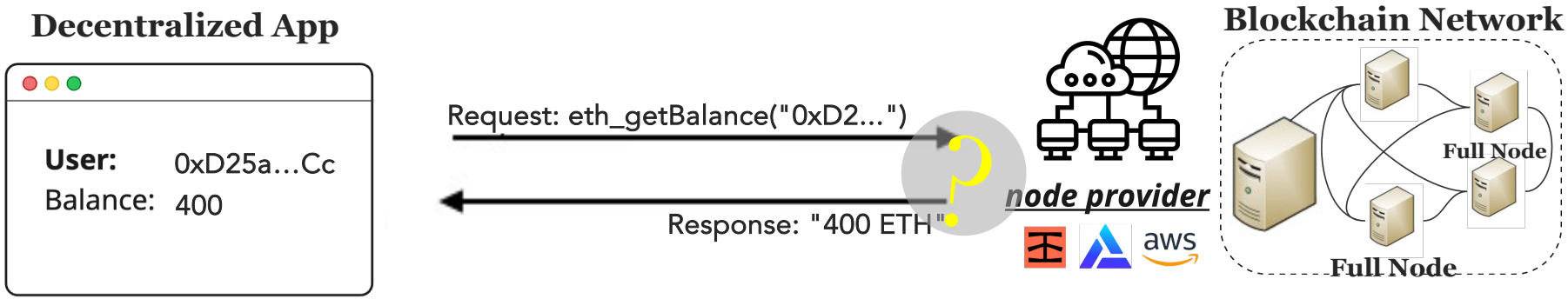




Requests in total: 400,000

2 USD

PARP brings back pseudonymity and data integrity



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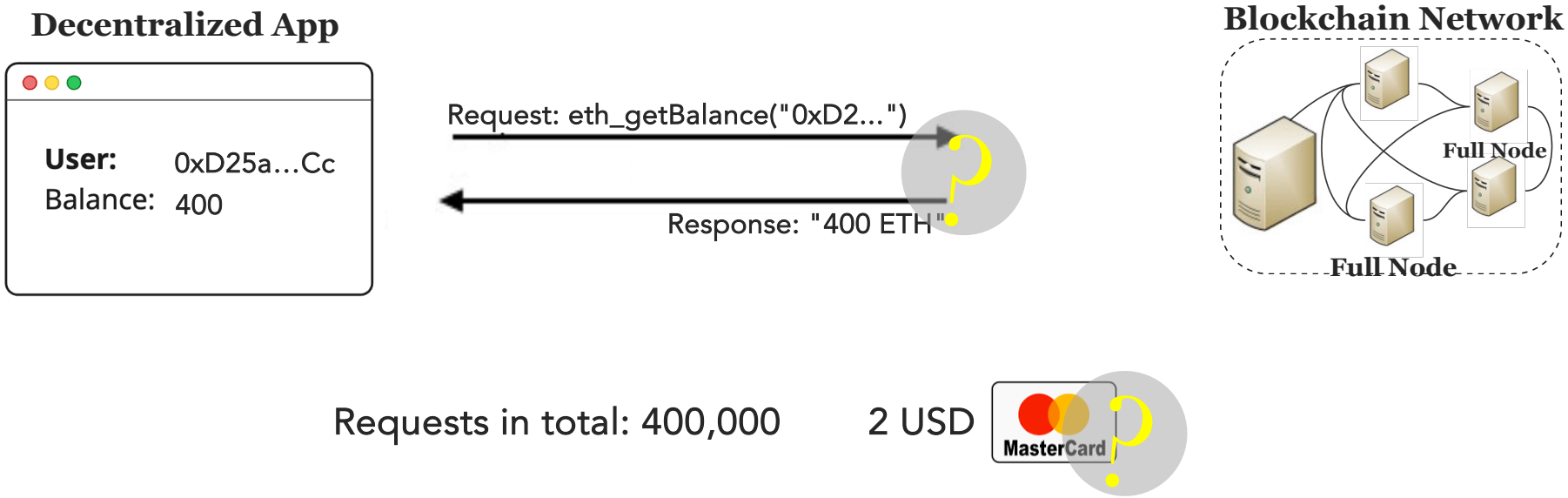

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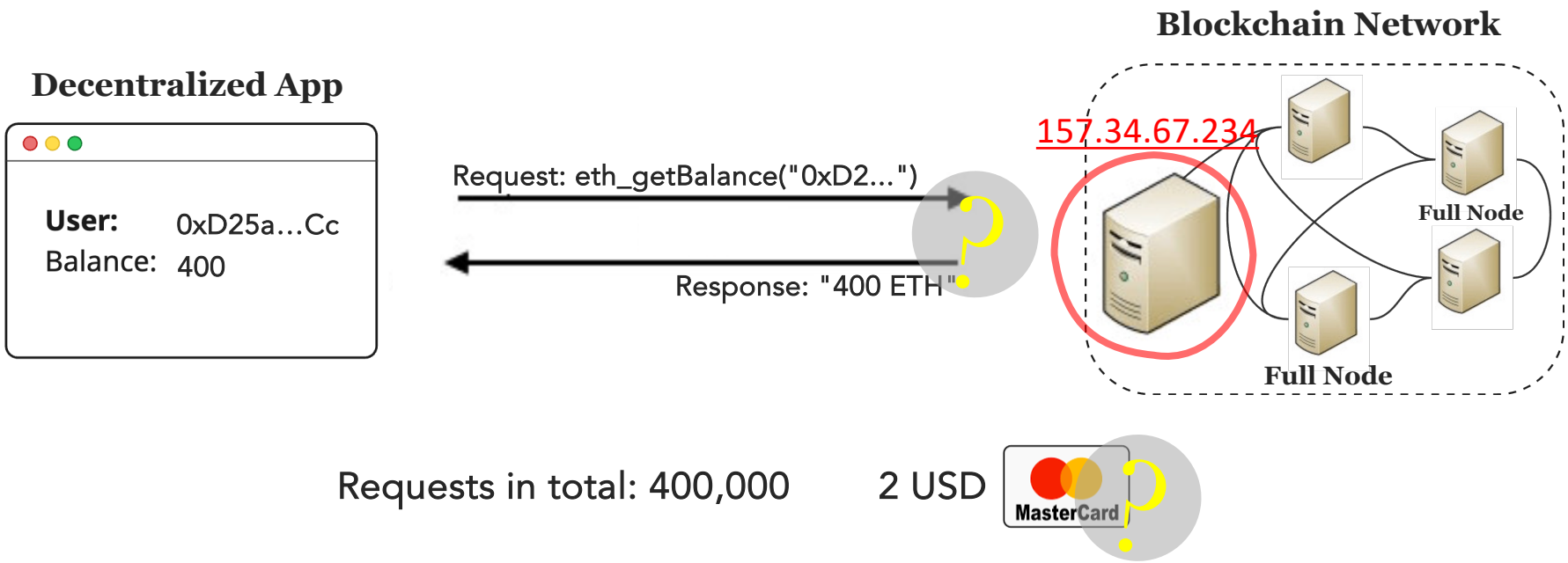

PARP brings back pseudonymity and data integrity

0xC8A7ae3f6Ae079c20BA191640
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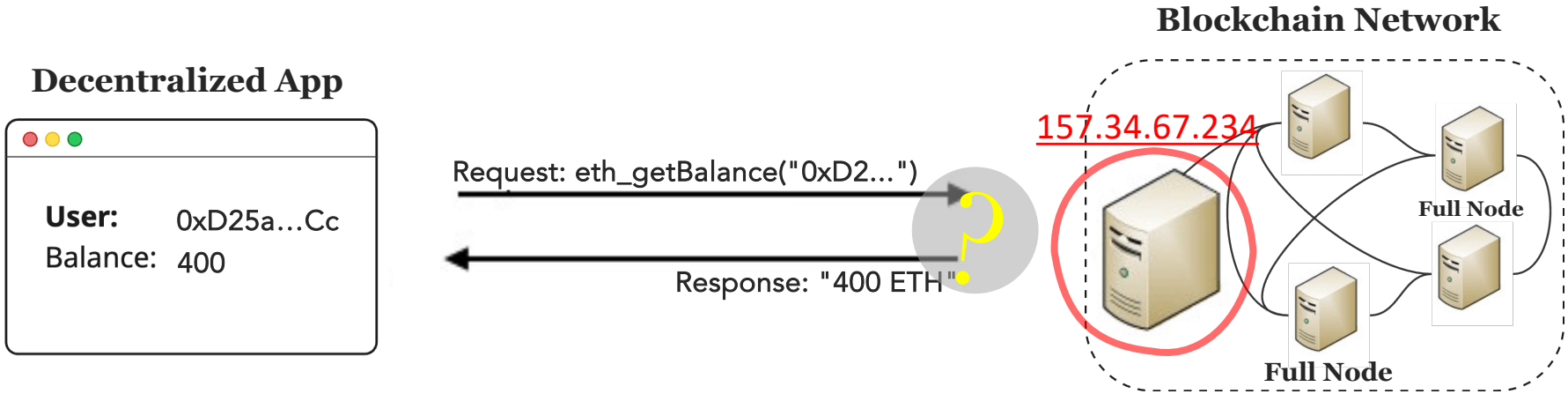
https://157.34.67.234:8888/API



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89143F48F7B965f

https://157.34.67.234:8888/API

Decentralized App

User: 0xD25a…Cc
Balance: 400

Request: eth_getBalance("0xD2...")

Response: "400 ETH"

Blockchain Network

157.34.67.234

Full Node

Full Node

Requests in total: 400,000

2 USD

MasterCard

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Light client

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Blockchain Network

157.34.67.234

Full Node

Full Node

Requests in total: 400,000

2 USD

I will pay you in cryptocurrency.

PARP – Overview

Light client

< 1GB

Only store block header info

Full Node

~ 1TB

PARP – Overview

Light client

< 1GB

Only store block header info

Full Node

~ 1TB

Assumption: Light clients have free access to block headers

Compact | **Not client-specific** | **Root of trust**

PARP Main Components

Light client

Full Node

PARP Main Components

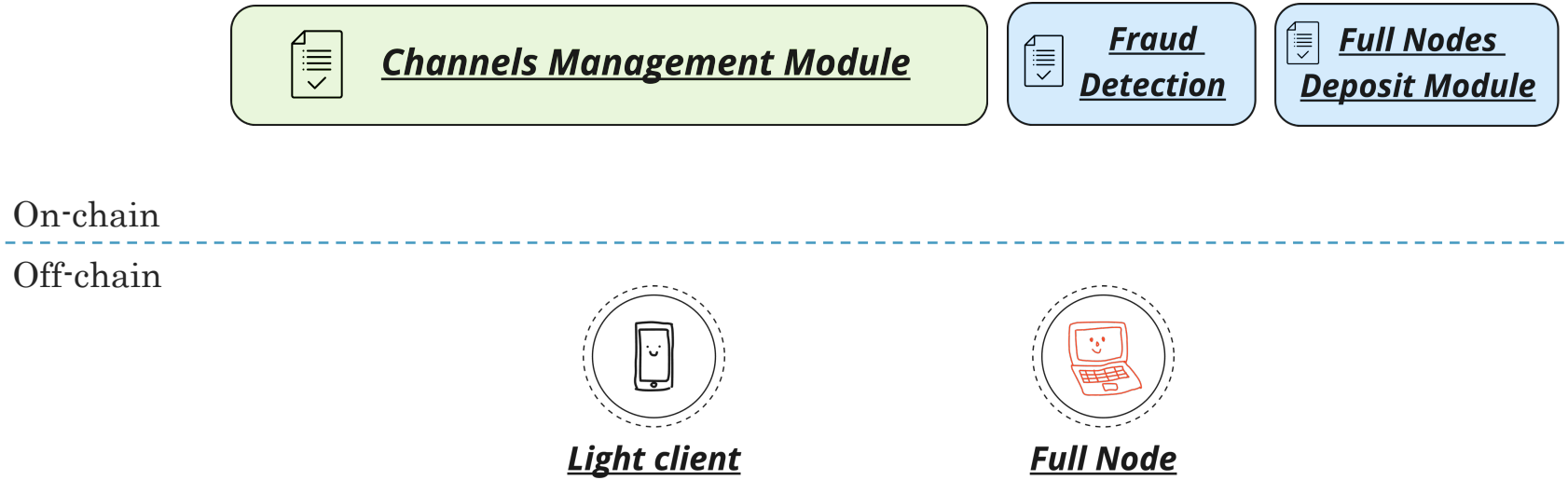
On-chain

Off-chain

Light client

Full Node

PARP Main Components



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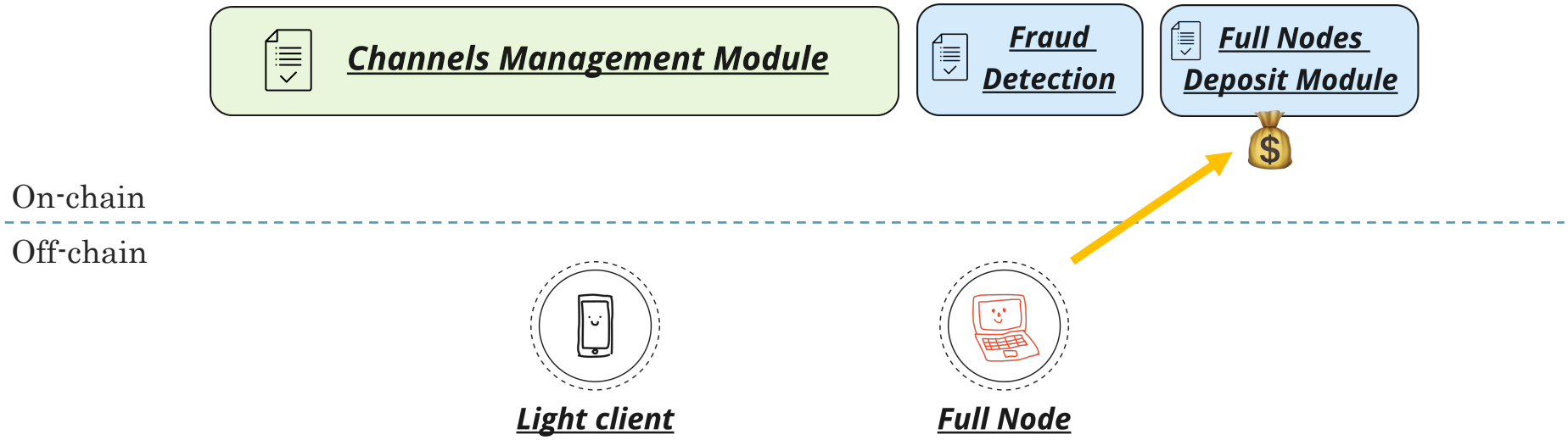
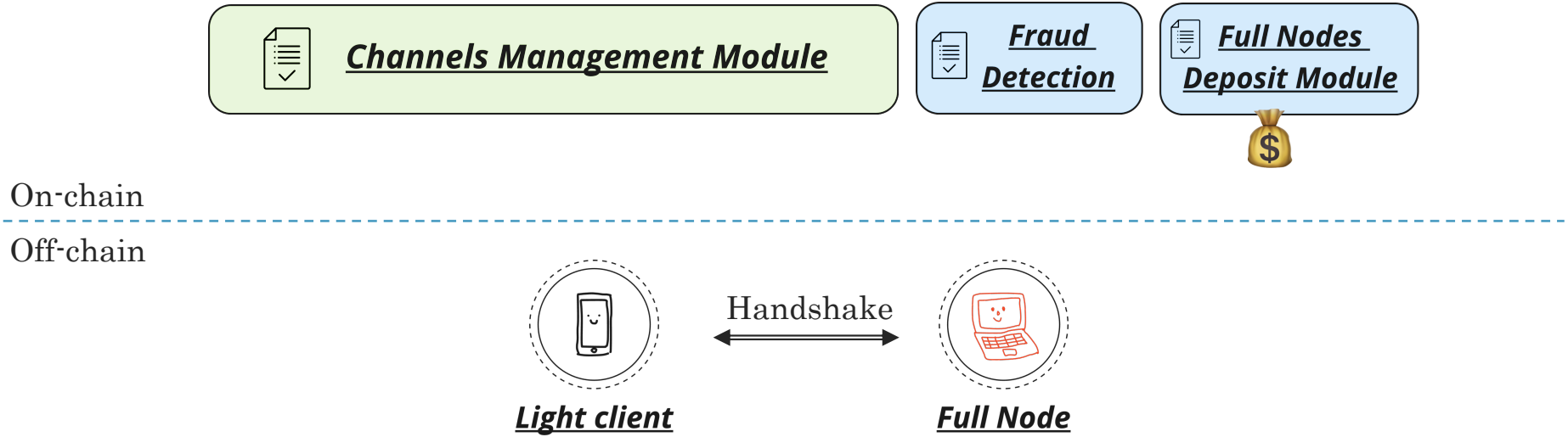


Full Nodes deposit first, then can accept connections.

PARP Main Components



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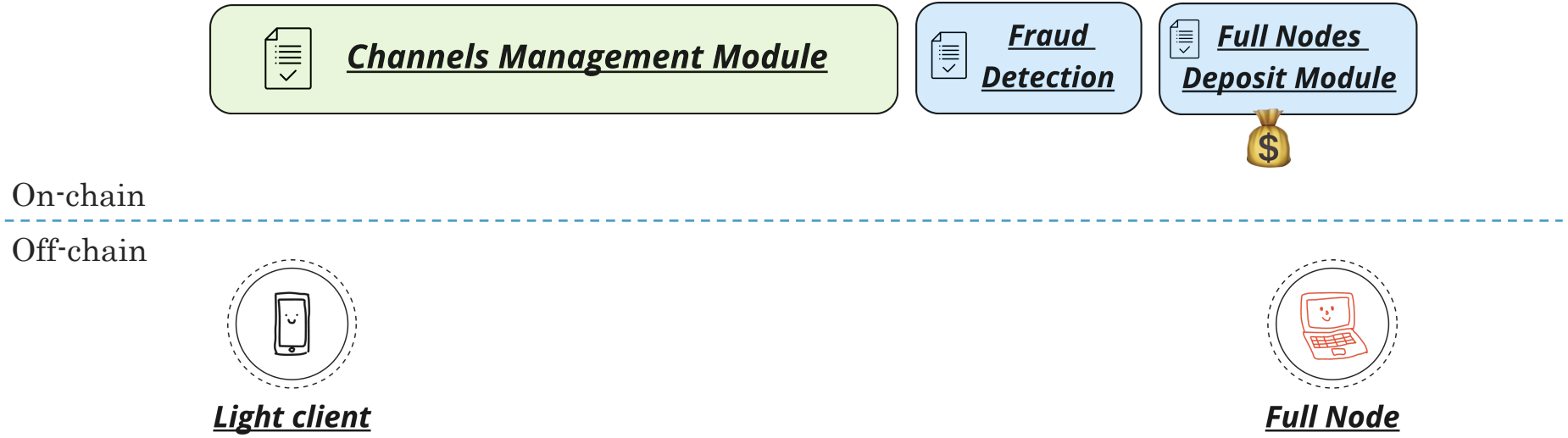
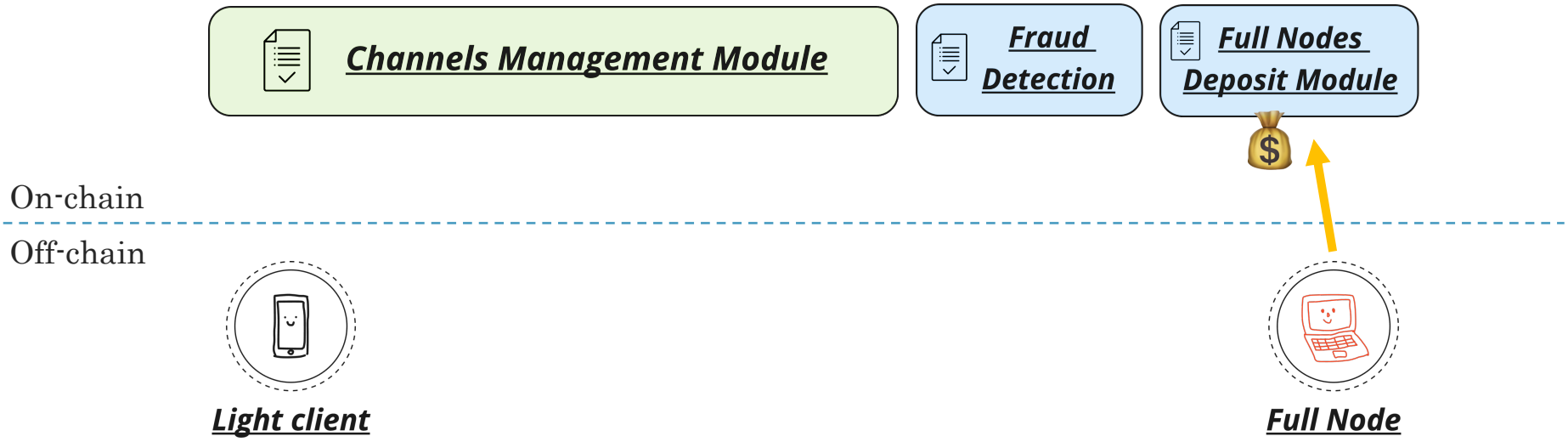


PARP – Active Phase



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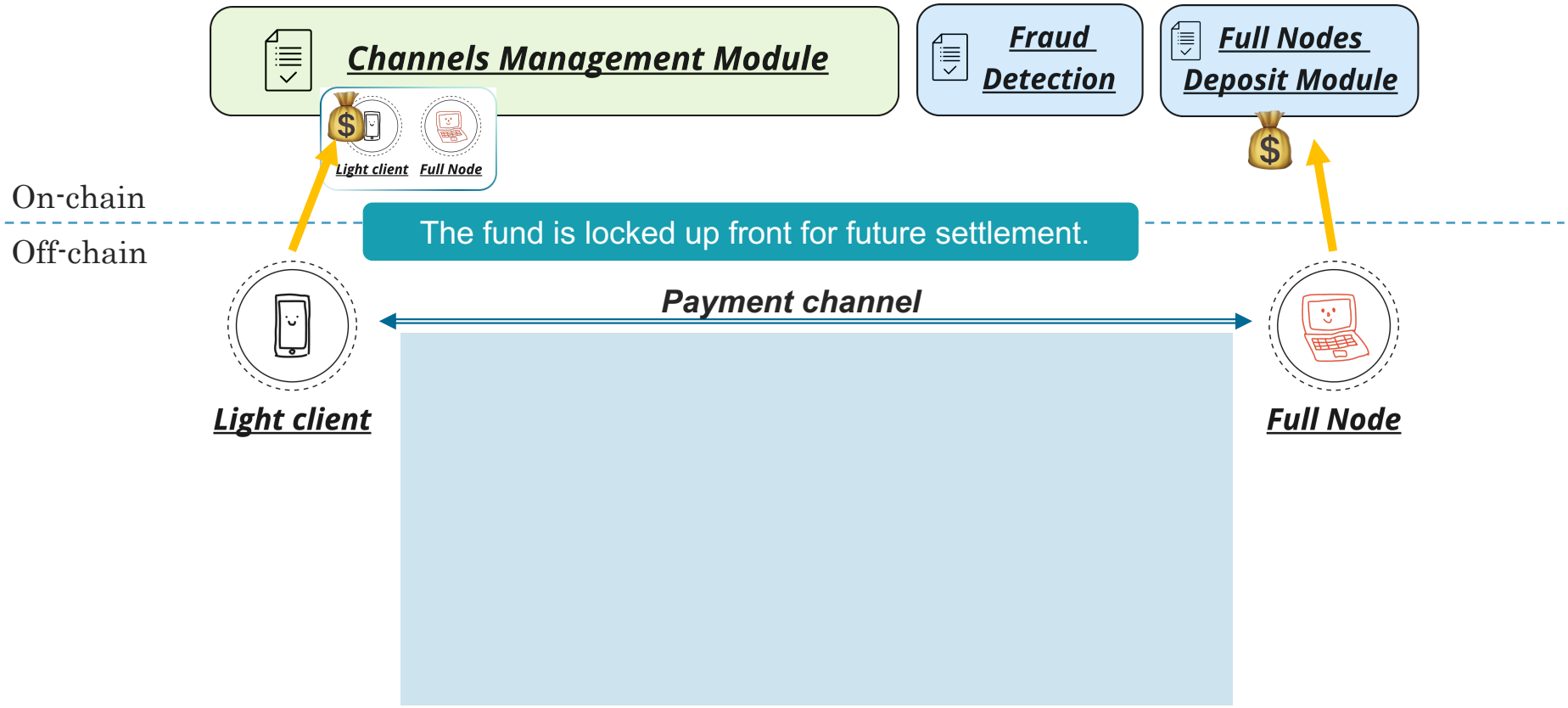

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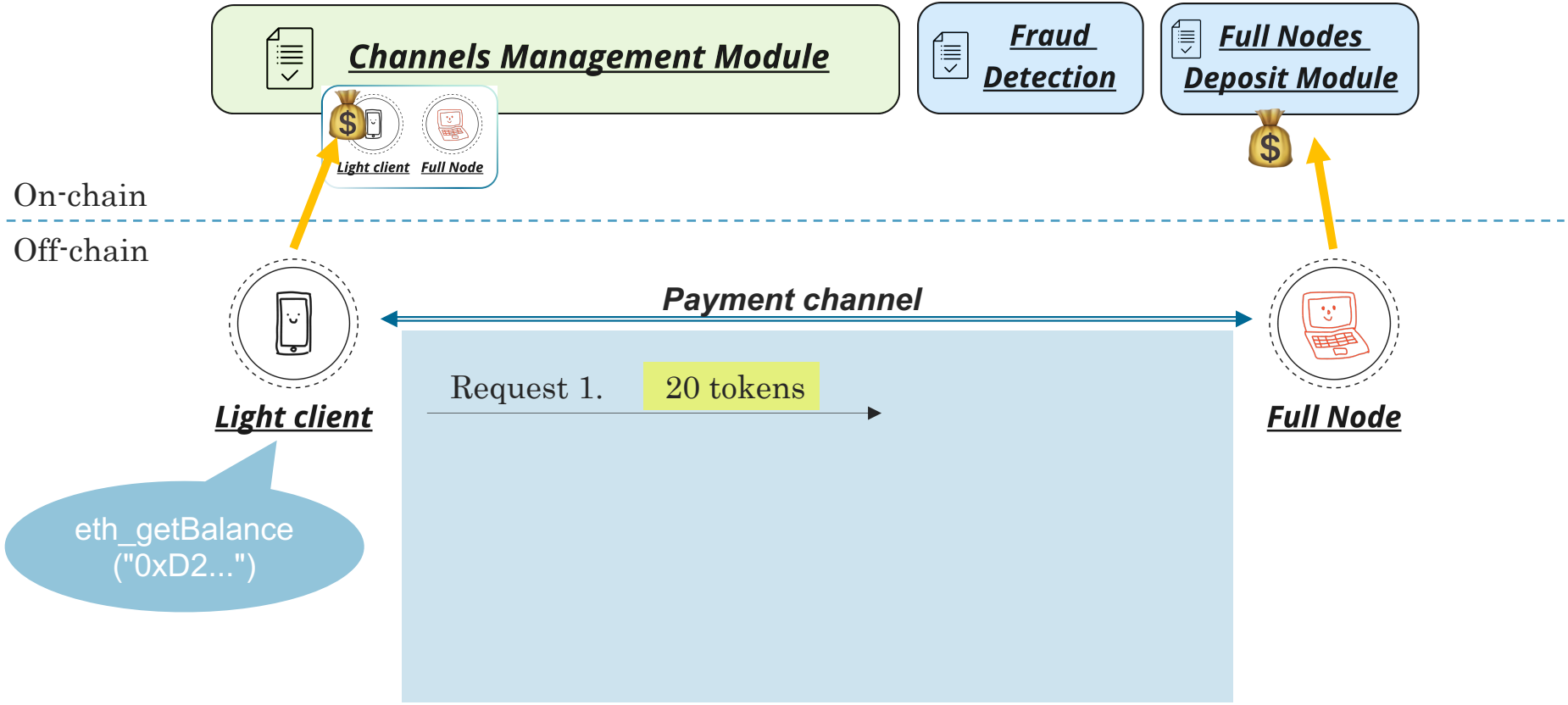

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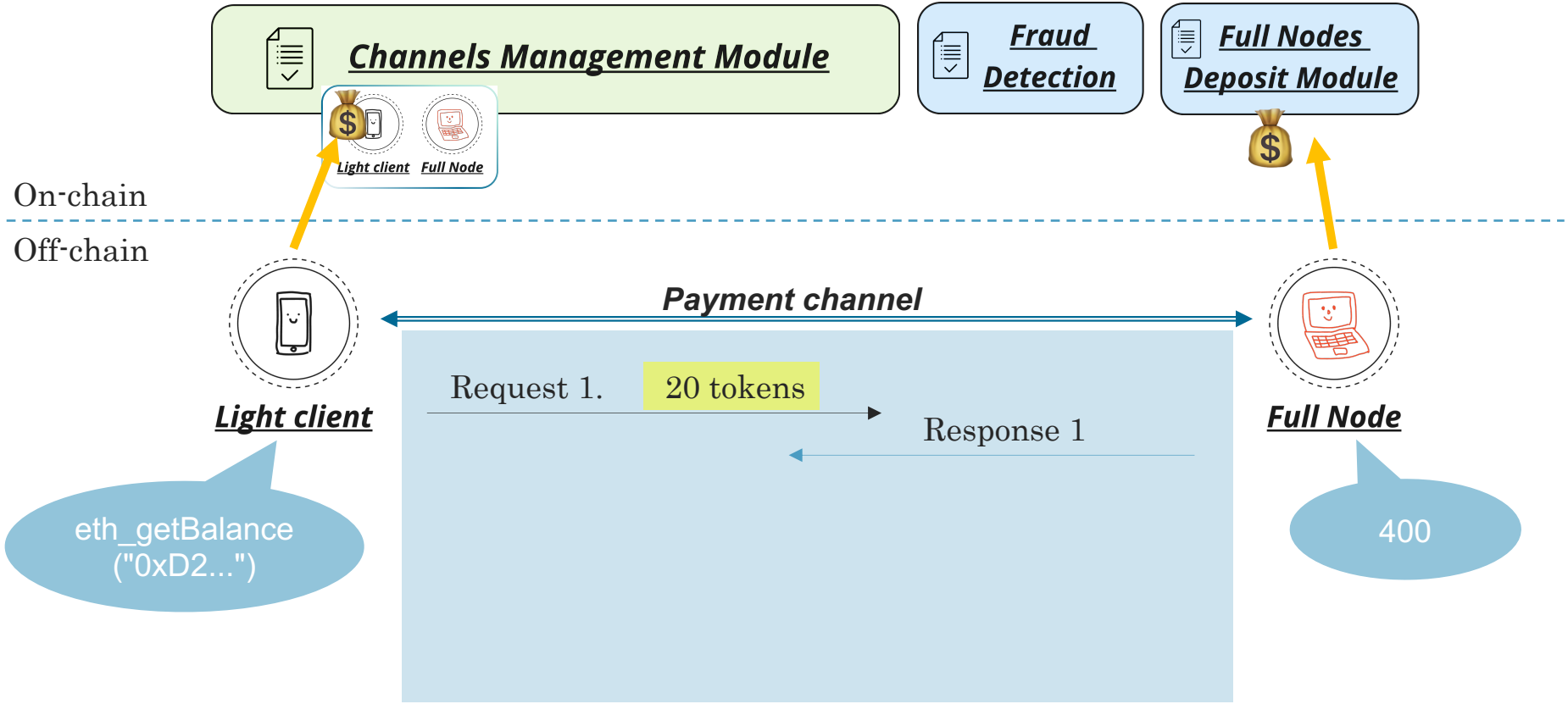

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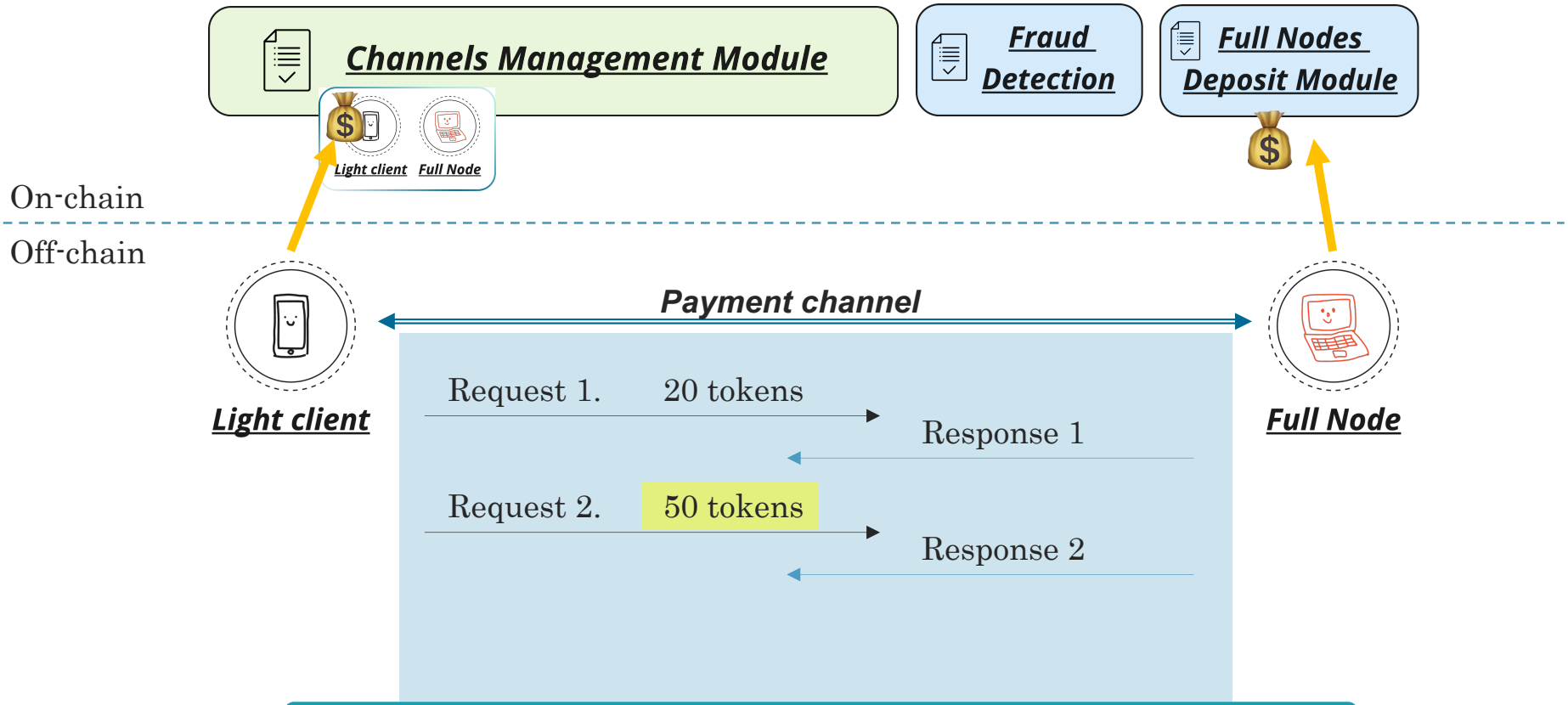
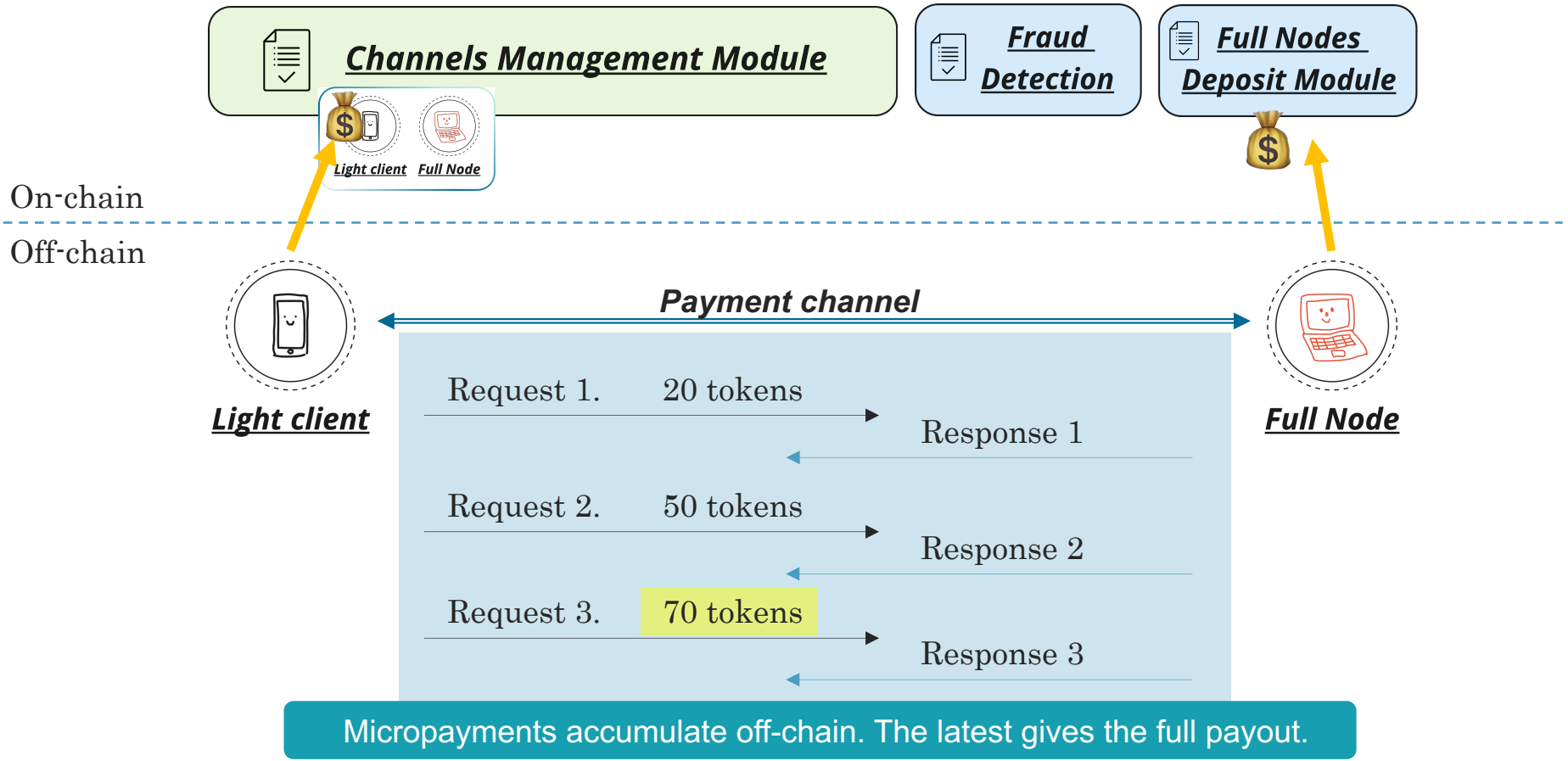
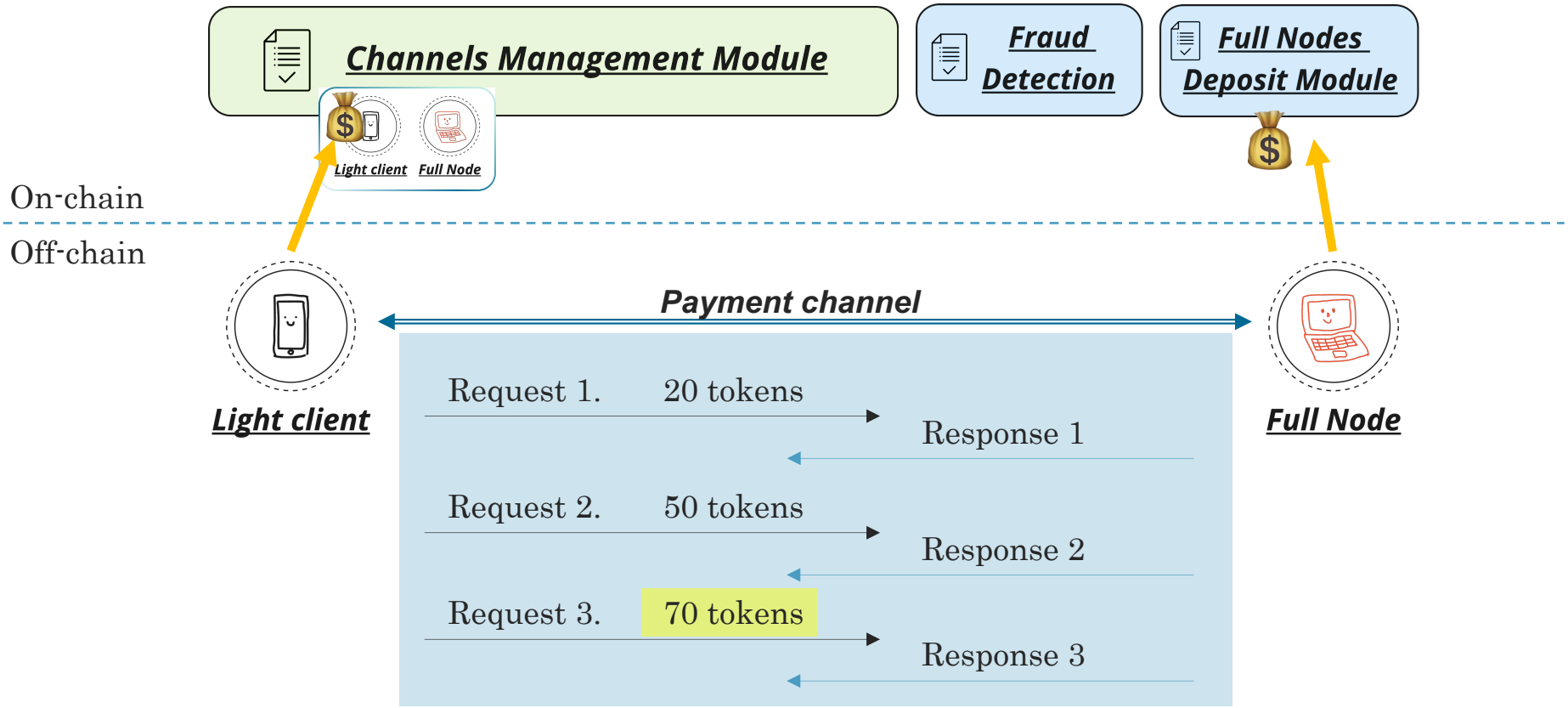


Micropayments accumulate off-chain. The latest gives the full payout.

PARP – Active Phase



PARP – Closing Phase



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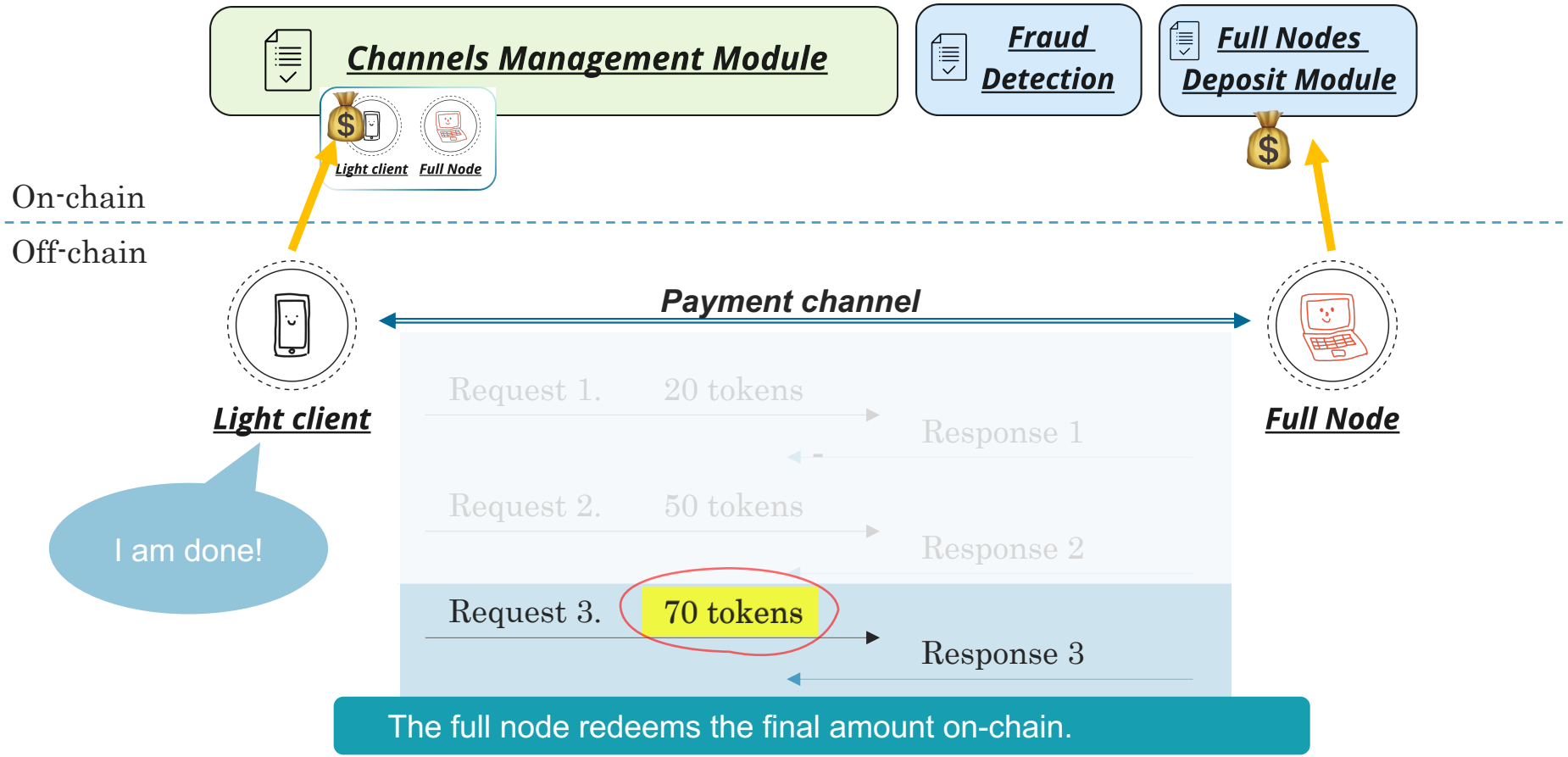

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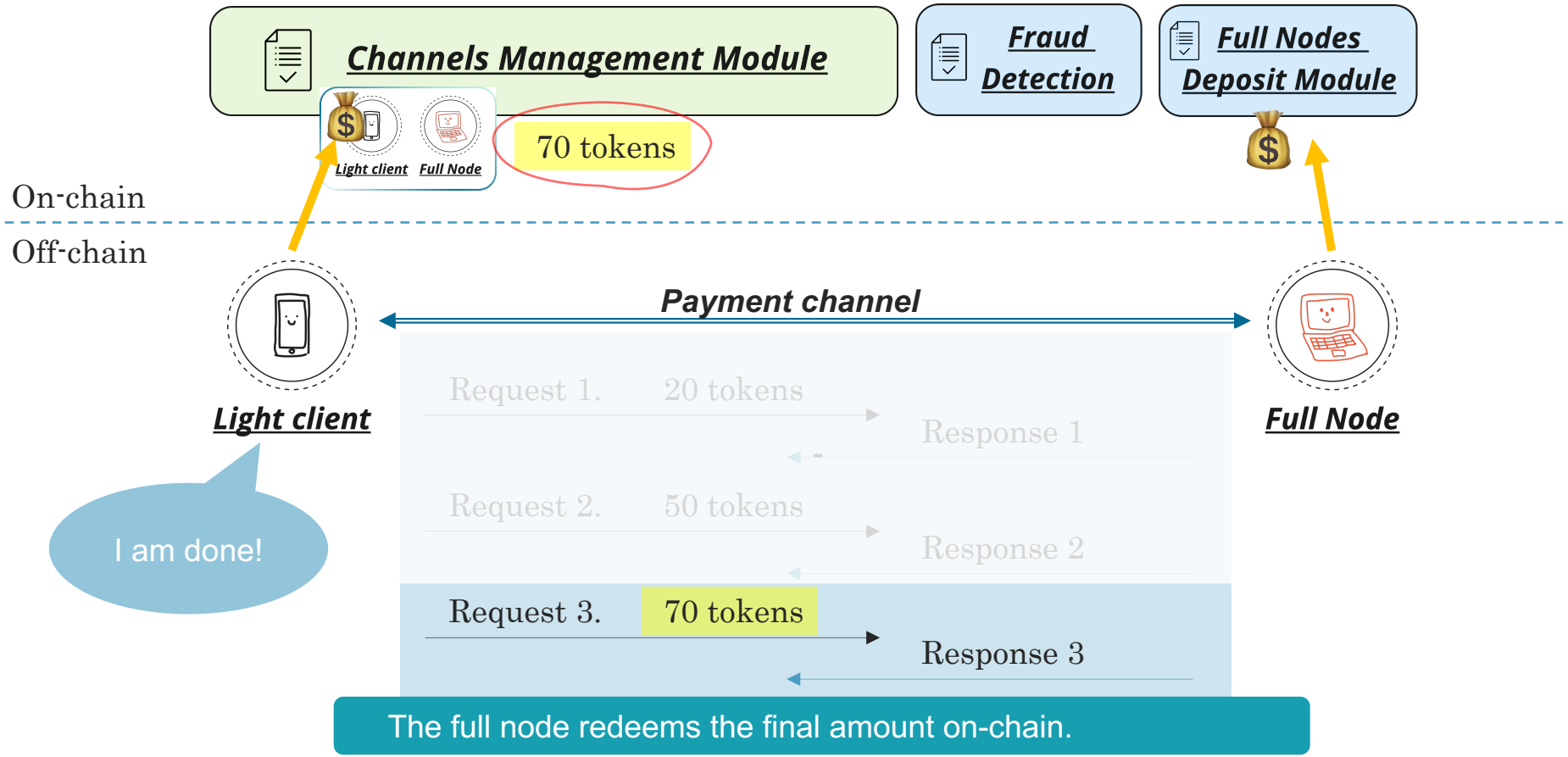

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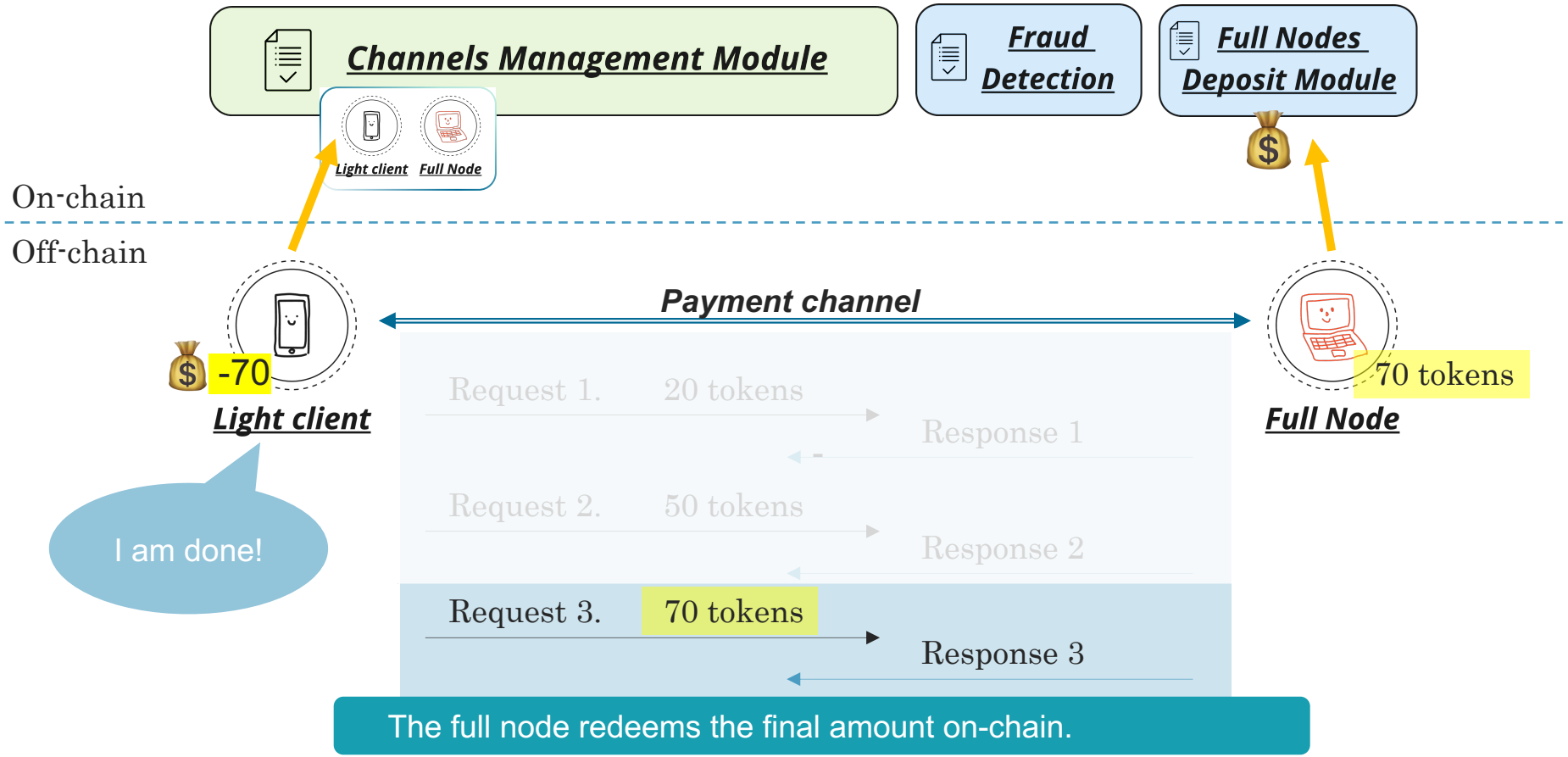

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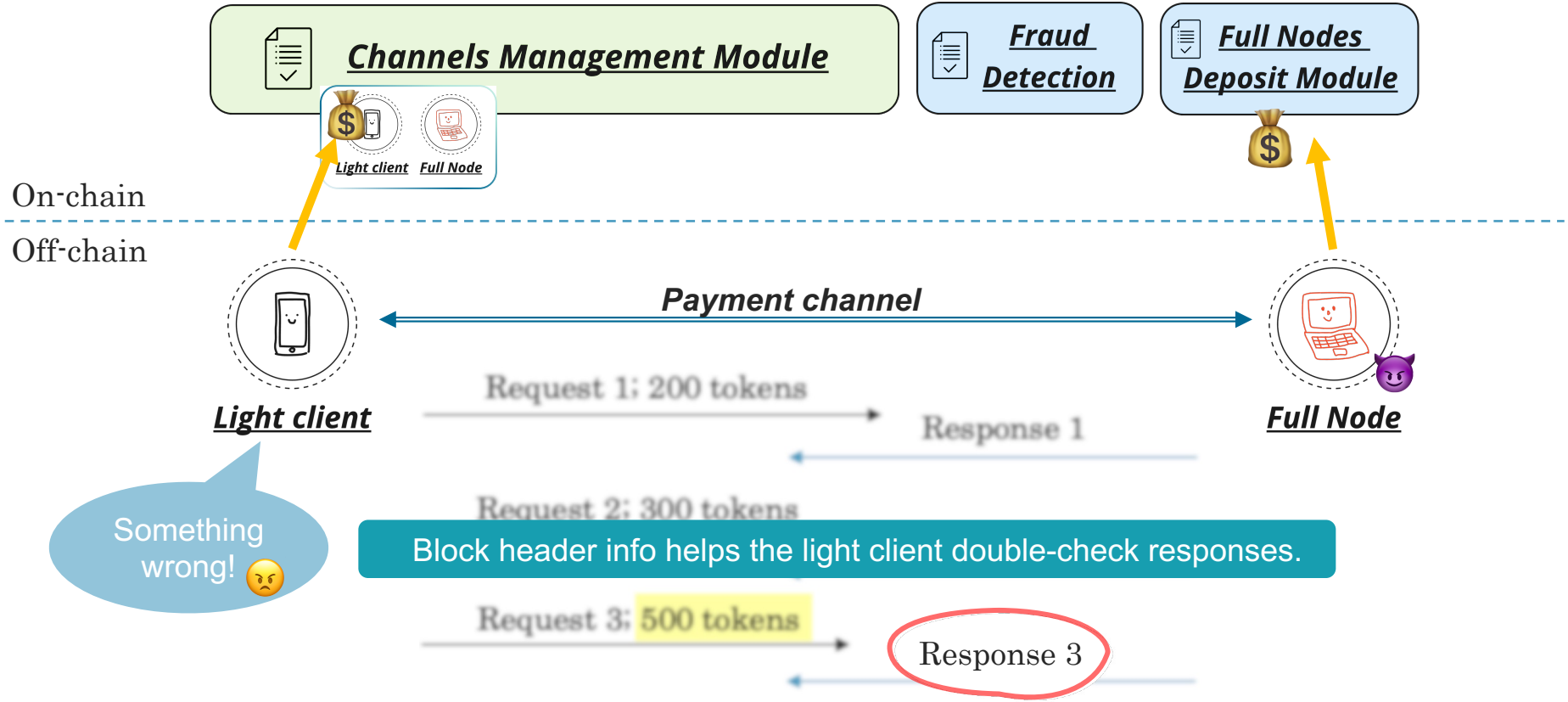

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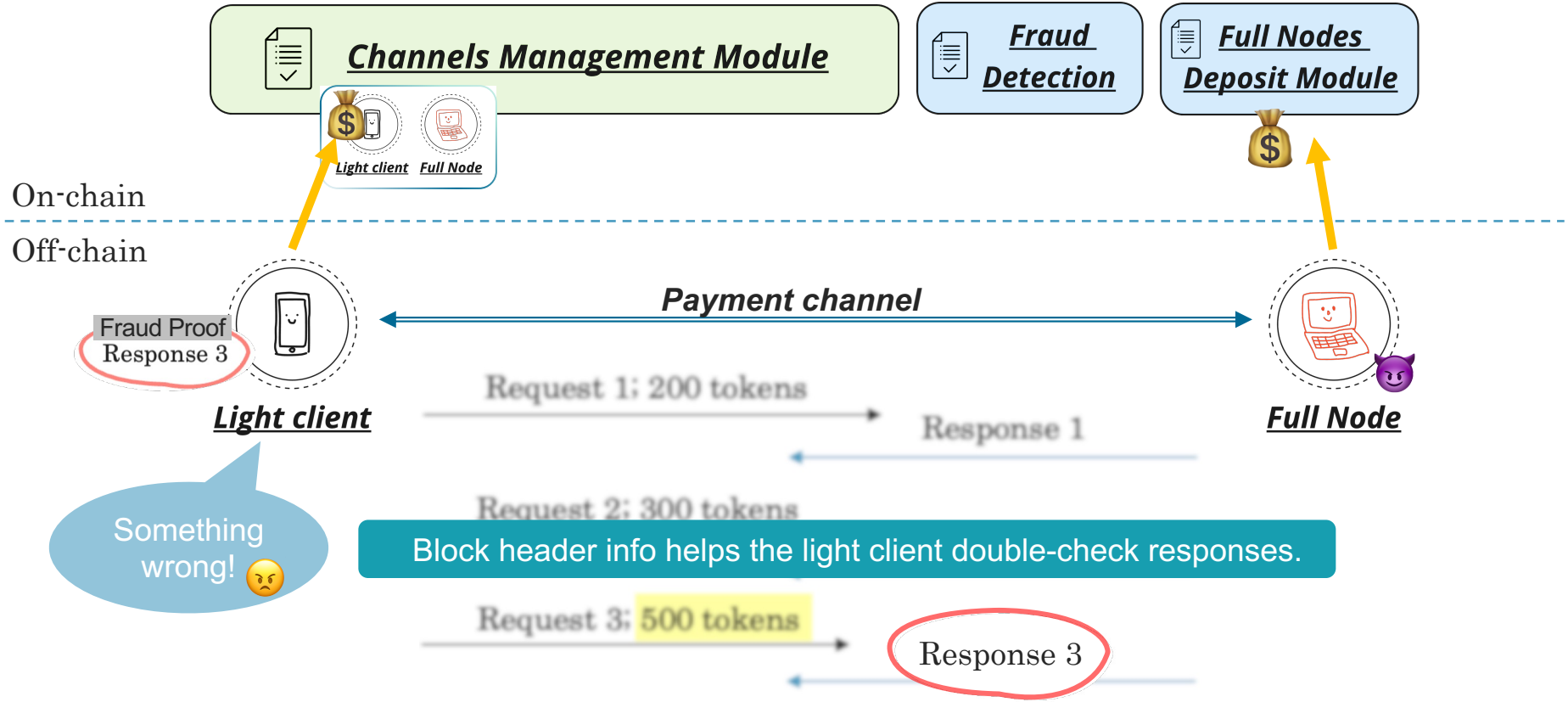

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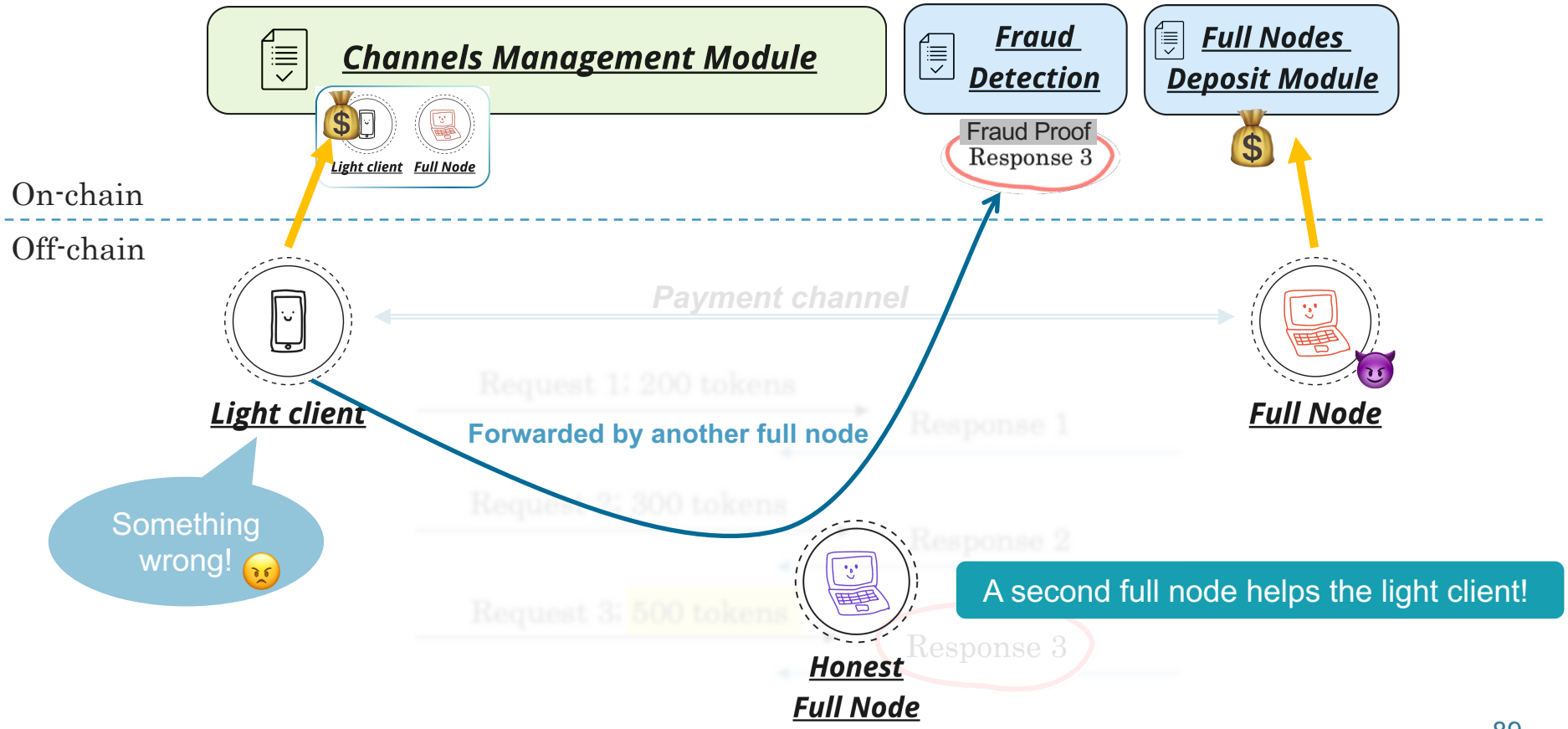

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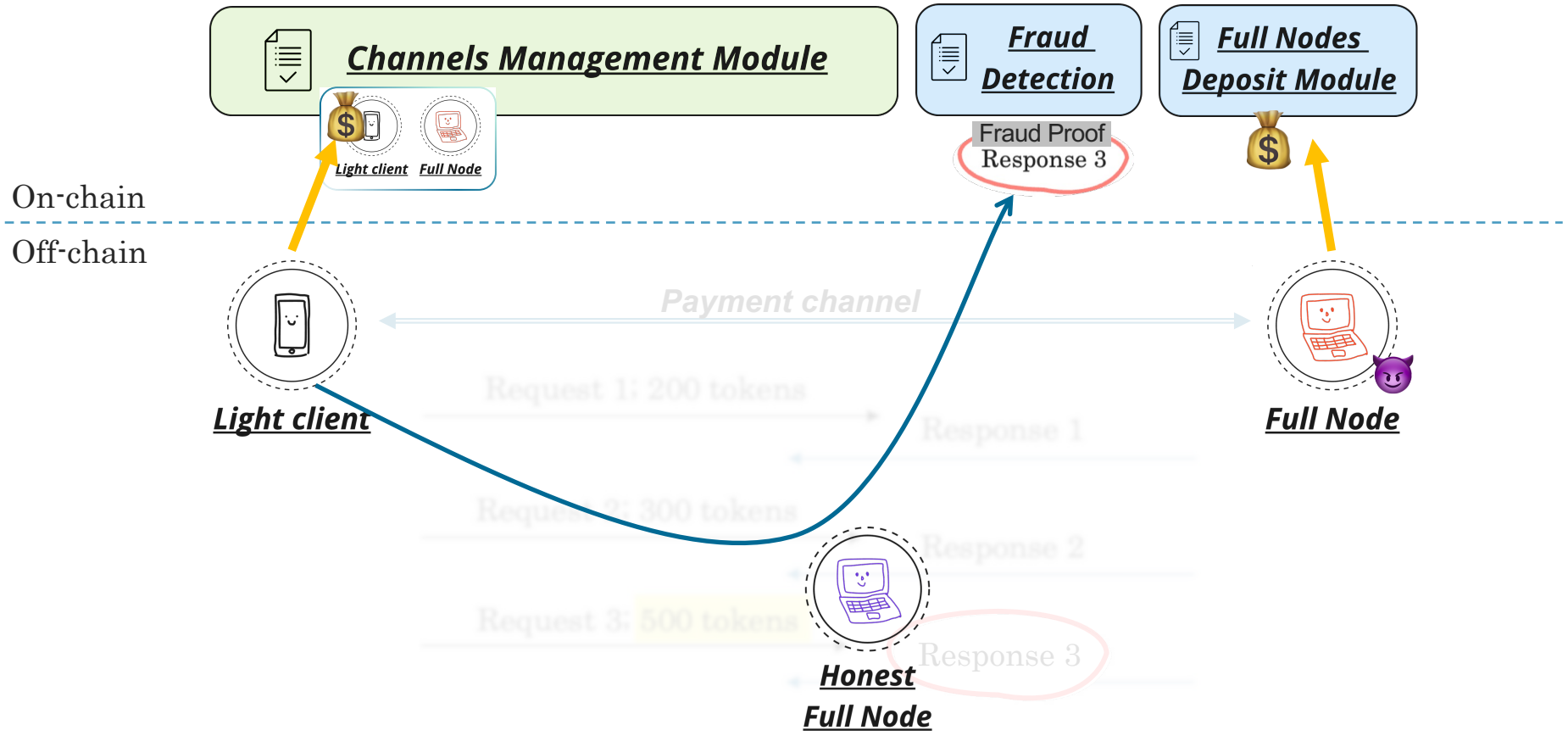

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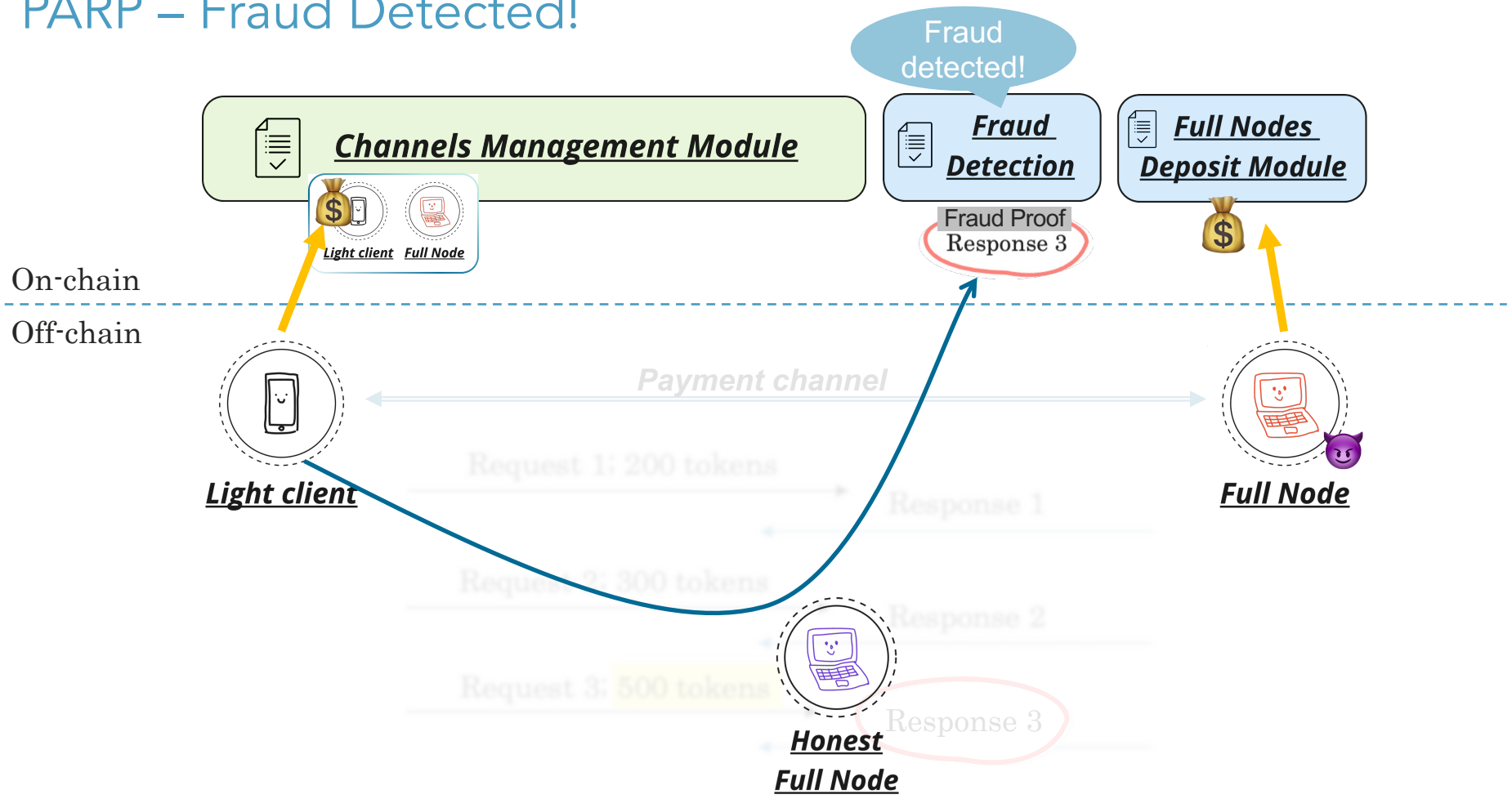

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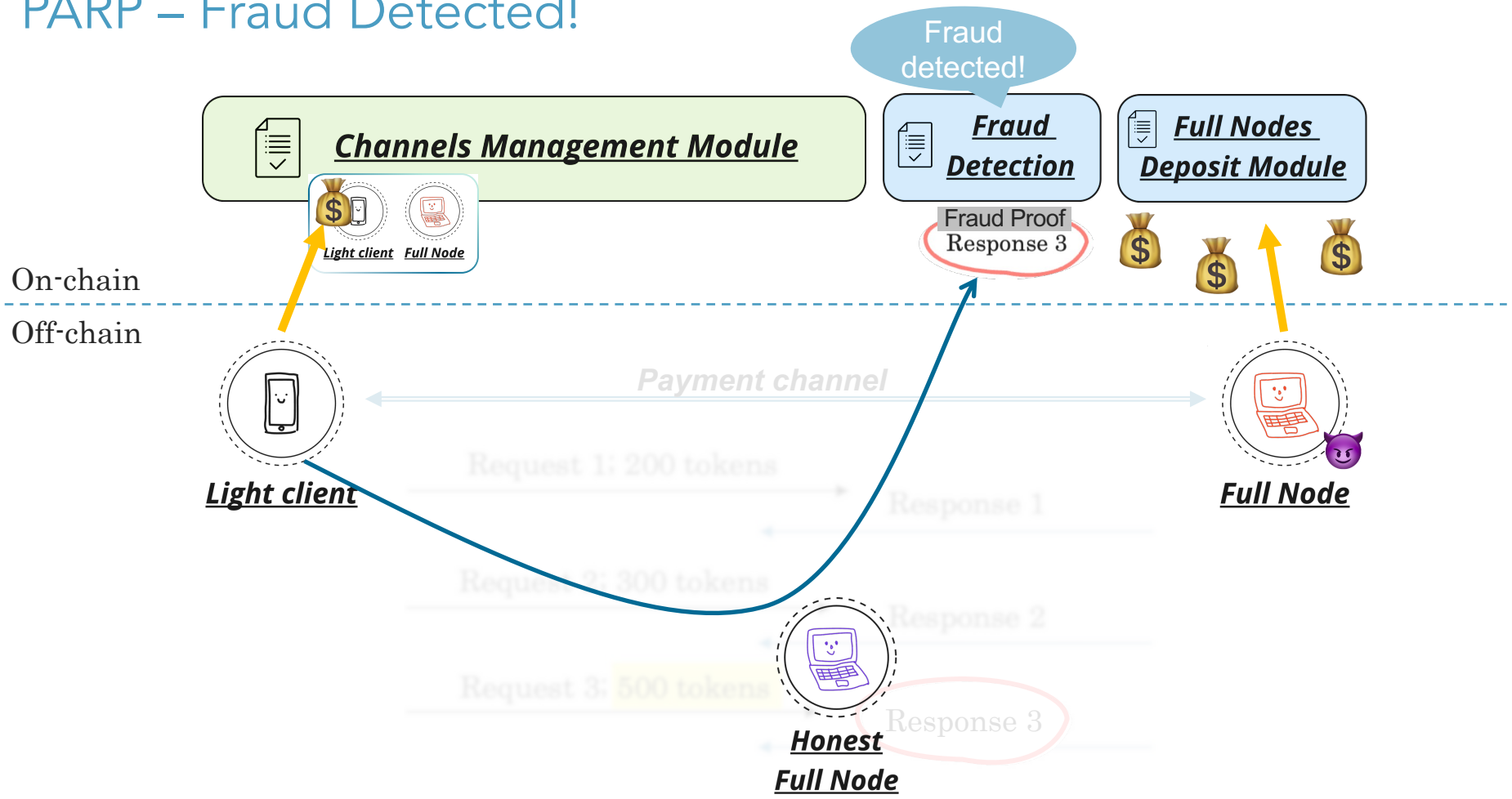

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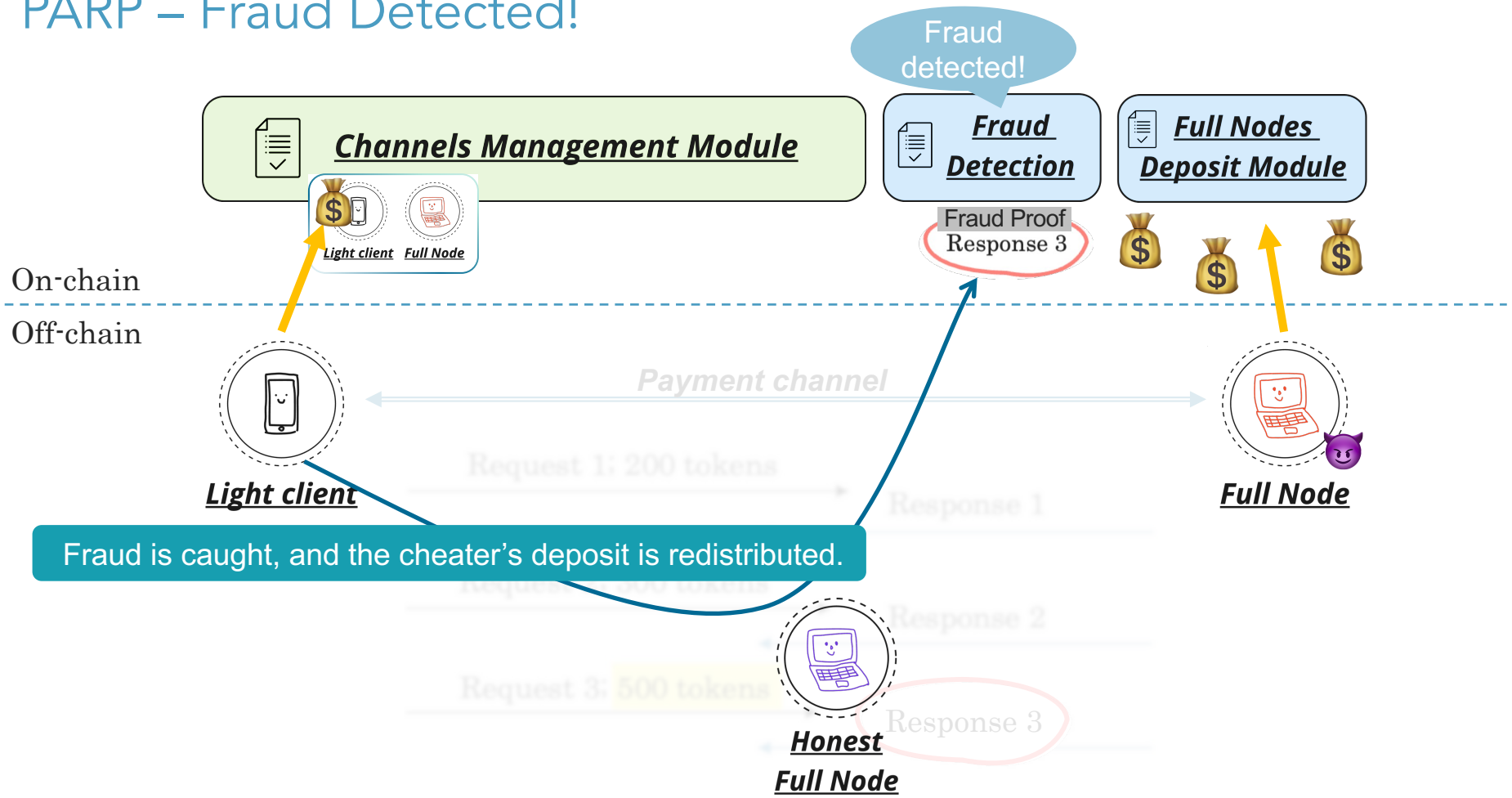

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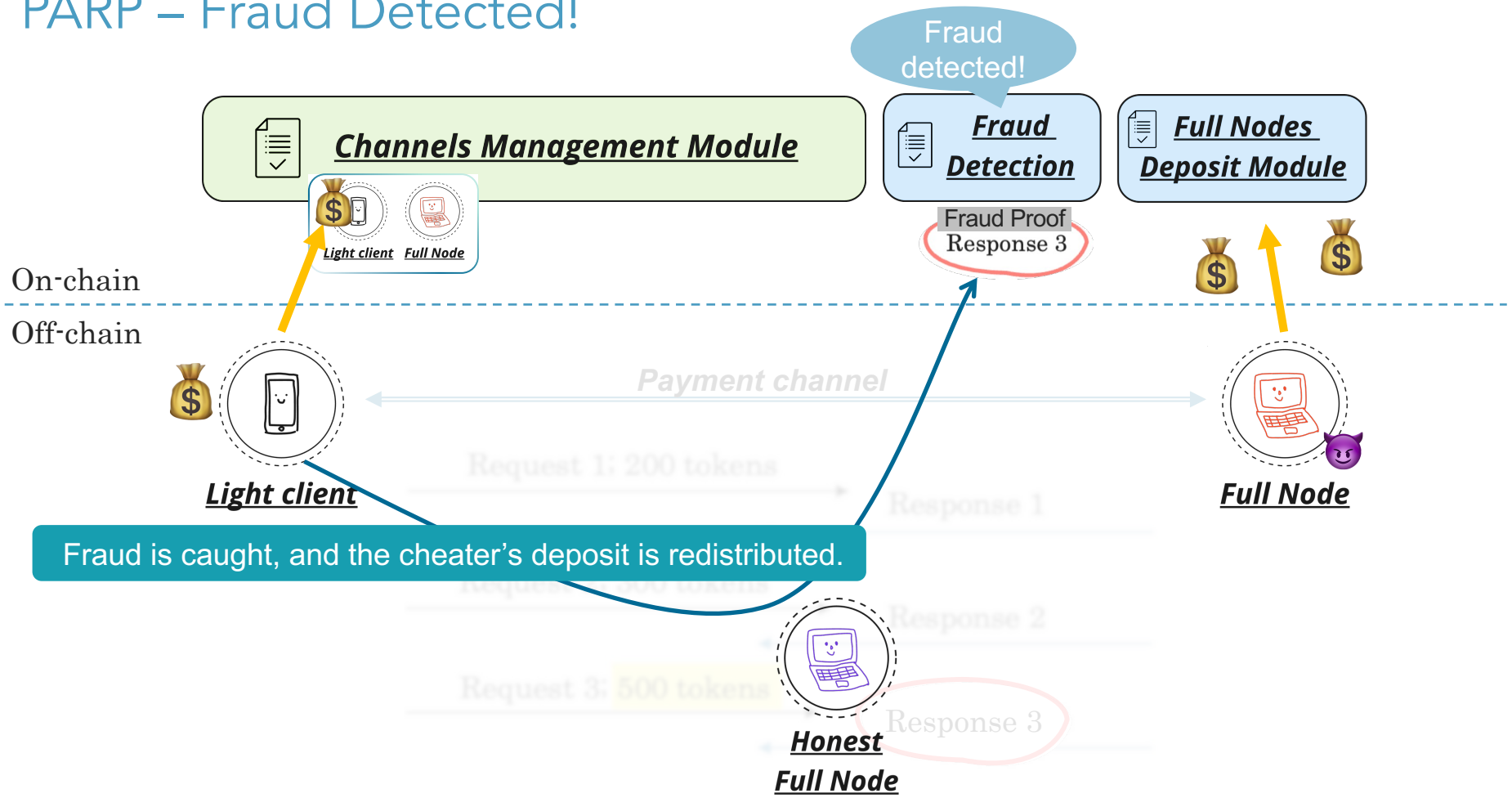

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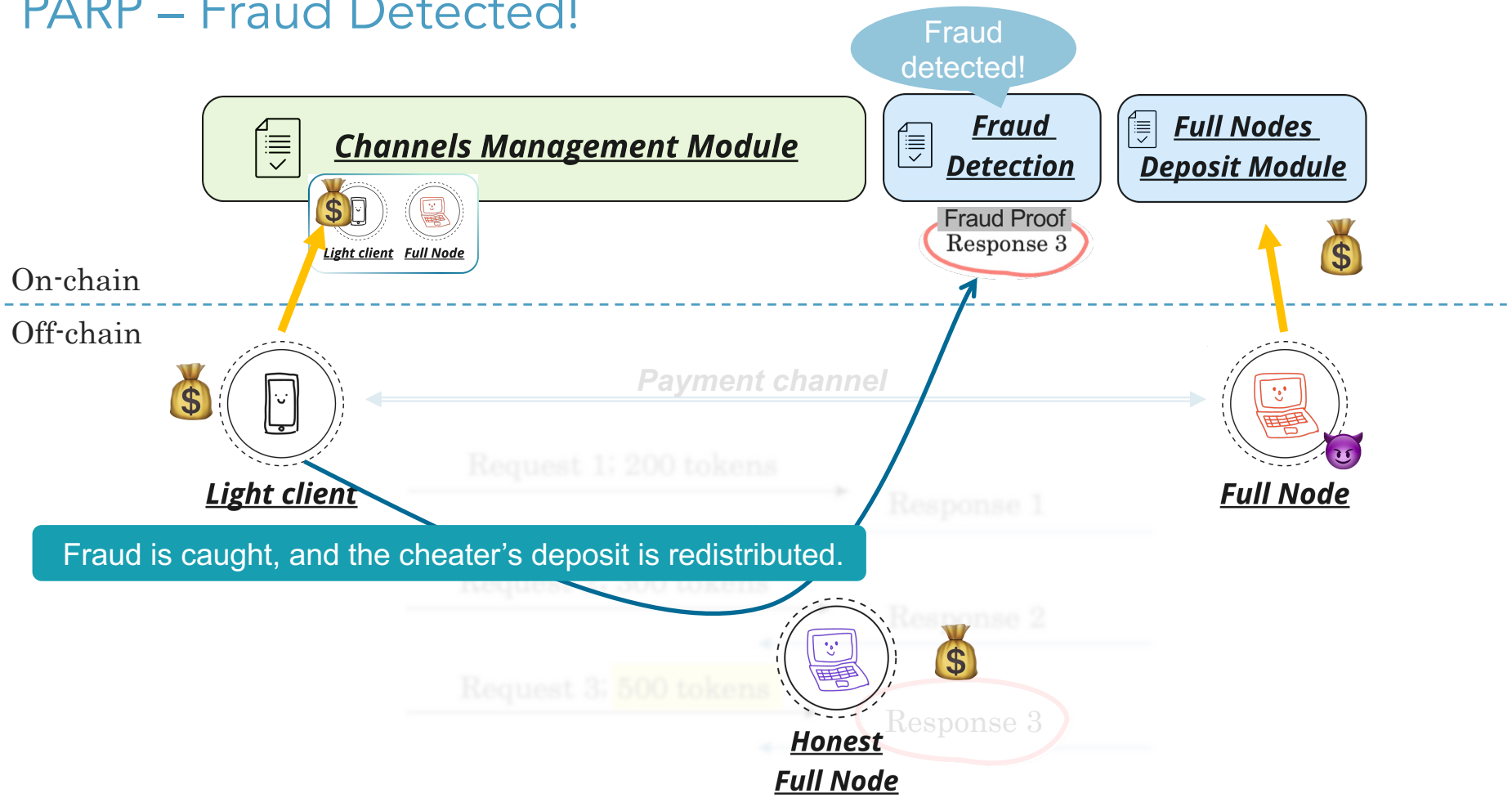

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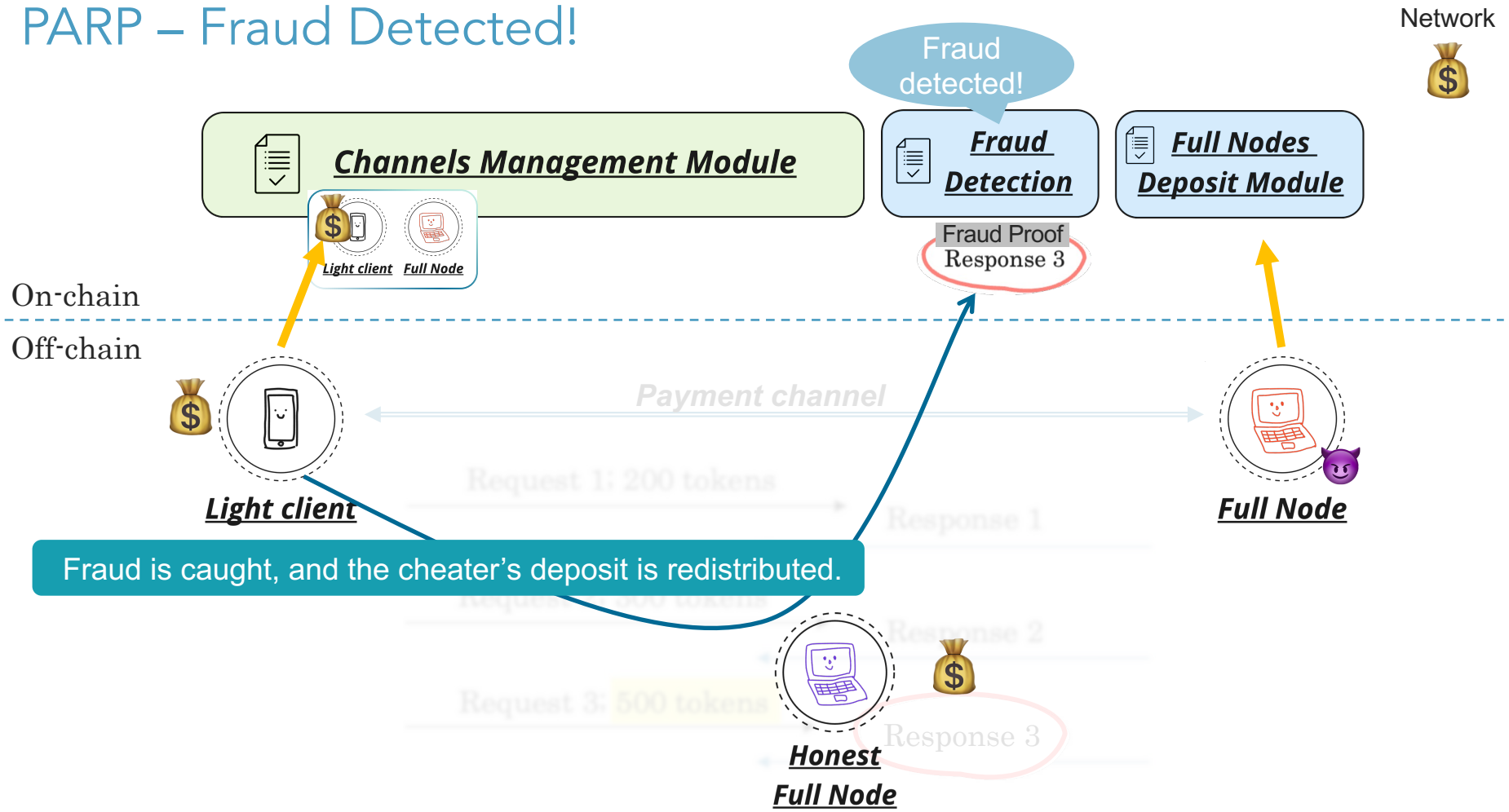

PARP Bridges Permissionless Access, Accountability, and Incentives

- **Permissionless**: no signup or gatekeeping, standard protocol
- **Accountable**: if a node lies, it can be caught and punished
- **Incentivized**: clients pay, nodes earn

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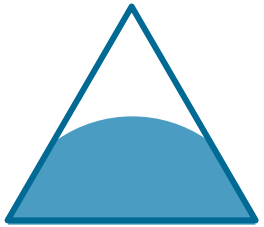
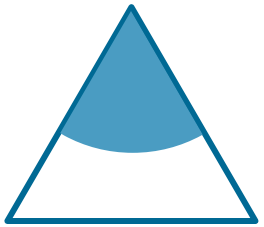


Node Providers



Public Endpoints

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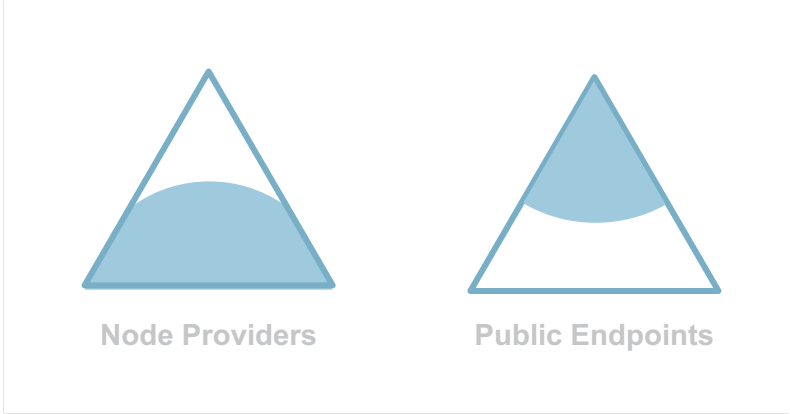
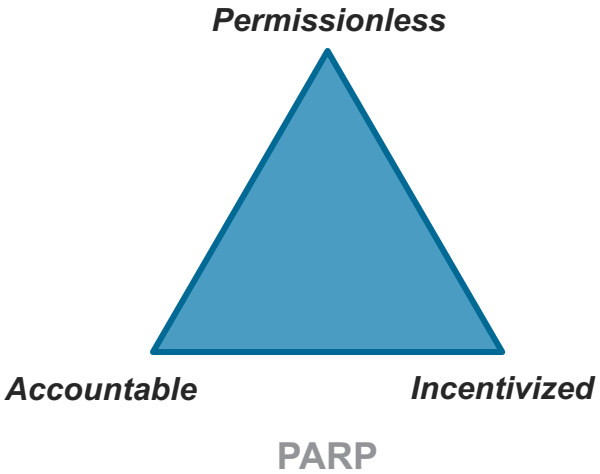
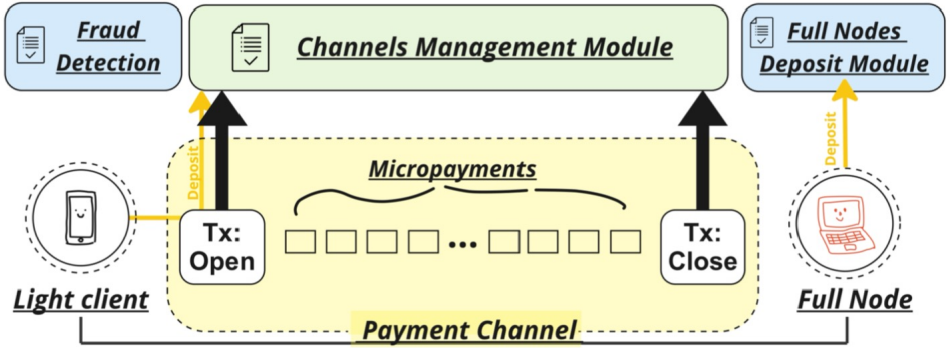




Implementation and Evaluation

Prototype was implemented on a local Ethereum network



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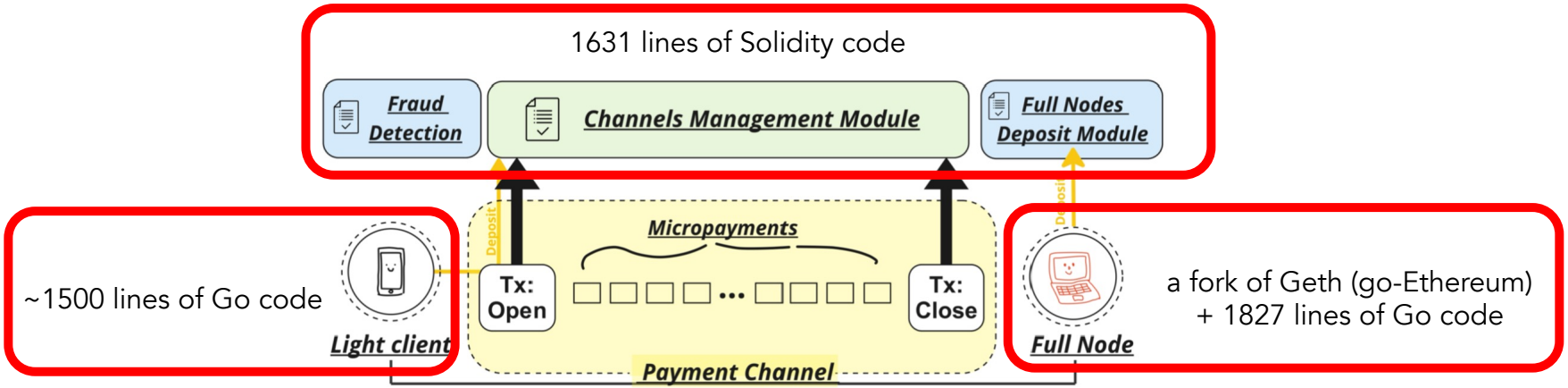

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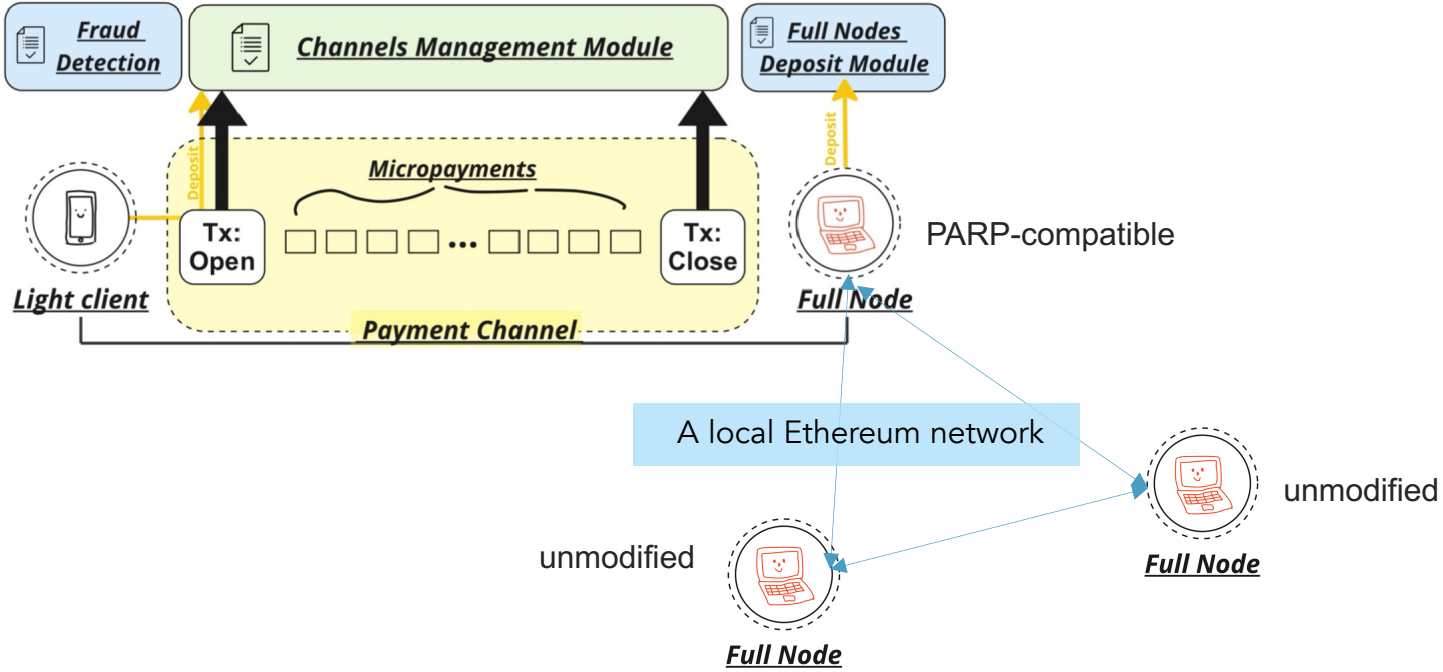

Evaluation details can be found in the paper

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	Size Overhead (in bytes)
PARP request	226 bytes
PARP response	187 bytes + Size of Merkle Proof

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Light Client Process Steps	**Write**	**Read**
(A) Request Generation	10.91ms	4.82ms
(D) Response Verification (proof)	7.13ms	5.78ms
(D) Response Verification (in total)	8.109ms	1.01ms
Full Node Process Steps	**Write**	**Read**
(B) Request Verification	714.43μs	703.13μs
(C) Response Generation (proof)	3.08ms	477.12μs
(C) Response Generation (in total)	3.37ms	1.29ms

Latency

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On-chain costs

Action	Gas Cost	MainNet (USD)	Arbitrum (USD)
Full Node Deposit			
Deposit funds	45238	2.171	0.018
Channel Management			
Open a channel	196183	9.417	0.078
Close a channel	110118	5.286	0.044
Confirm closure	87128	4.182	0.035
Fraud Proof Detection			
Submit a fraud proof	762508	36.6	0.305

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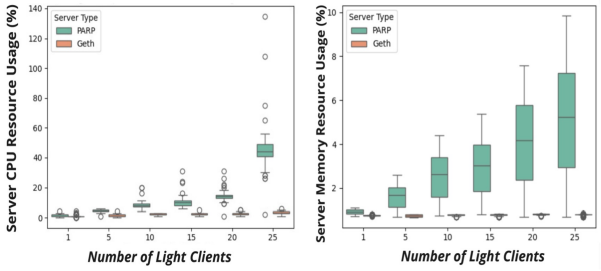
CPU & memory usage



Future work

- Minimizing the on-chain costs and scalability

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- Minimizing the on-chain costs and scalability
- Risks of single-node dependence
- Privacy concerns in full node interactions
- Formalization of cryptoeconomic incentives
- Network-level rewards

Get yourself a **permissionless** and **accountable** connection to a decentralized world.

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Depermissioning Web3: a *P*ermissionless *A*ccountable *R*PC *P*rotocol for Blockchain Networks

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